

VANDERBILT UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER
ENGINEERING

ECE 4380 Design Project

QRP Transmission Amplifier

Design & Simulation Report

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Executive Summary

We were tasked with designing an amplifier for a QRP Texting Transmission circuit involving a NXP **SA612** and 75Ω antenna. Our amplifier constraints were designed around these components and the QRP protocol. Our amplifier had to satisfy the following major constraints:

Circuit Requirements

- $V_o/V_i = 60 \pm 1.5$ dB, which is 1000 V/V. (High)
- Broadband width is from 0.5 to 7.2 MHz.
- $R_{in} = 1.5 \text{ k}\Omega \pm 20\%$ (Moderate)
- $R_{out} = 75 \text{ }\Omega \pm 20\%$ (Low)
- Power = $\leq 12\text{mA}$ from +9V supply
- Survive $\pm 50\%$ β_o Variations

It can be noted that the input and output specifications exist to impedance match the amplifier with the SA612 which features a $1.5 \text{ k}\Omega$ output impedance and the 75Ω antenna.

To meet these requirements we were provided six possible BJT transistors, resistors, and capacitors. A power fee was also implemented above 60 mW at a rate of 15 cents per mW.

To design an amplifier that meets these requirements, we designed a six stage BJT amplifier featuring a darlington input stage, followed by an emitter follower buffer, two cascode gain stages separated by another emitter follower buffer, and a darlington output stage. We introduced shunt feedback to our input stage to moderate our R_{in} value and series-shunt feedback to stabilize our gain stages and keep them in spec across the target frequency band.

Our design successfully met all criteria, achieving a simulated midband gain of 60.6 dB, a bandwidth of 12.3 MHz, with R_{in} and R_{out} values falling within $1.5 \text{ k}\Omega \pm 20\%$ and $75 \text{ }\Omega \pm 20\%$. We also hit a maximum power consumption rate of 60.5 mW ensuring that we avoided any price penalty and stayed below the imposed 12mA hard limit.

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1 Introduction / Background / Scope / Motivation

QRP radio transmission is a communication protocol within the amateur radio community involving maximum transmission range and informational content using minimal power. This radio design philosophy is based on the U.S. FCC Part 97 rule mandating that amateurs should “use the minimum power necessary to carry out the desired communications.”

Texting via QRP involves using **PSK31** digital encoding of text information within an amateur radio frequency band. To transmit a message, PSK31 is used to convert binary inputs to an analog waveform and the computers sound card is used to create an audio wave train of the binary sequence by imparting to an audio tone where 180° phase shift represents a 0 and no phase shift represents a 1. A typical radio frequency (RF) transmission band used for this purpose is the high frequency (3-30MHz or ITU band 7), specifically the 40-meter wavelength, or 7.1MHz. We were tasked with designing an amplifier for a QRP Texting Transmission circuit involving a NXP **SA612** and 75Ω antenna.

The goal of the design project is to meet the criteria defined in the highlighted box. The expected input signal for the amplifier originates from the SA612A mixer/oscillator integrated circuit, which will deliver a modulated 7.1 MHz signal to our amplifier that contains audio wave from the PSK31 converted signal. The SA612A has an output impedance of $1.5\text{ k}\Omega$, which is responsible for the design specification of the amplifier input resistance being equal to $1.5\text{ k}\Omega \pm 20\%$ over the entire bandwidth of operation. The output of the amplifier will drive a $75\text{ }\Omega$ coaxial cable to an antenna with minimal power loss and reflections. This is maintained by the specification of a strict output impedance of $75\text{ }\Omega \pm 20\%$ over the entire bandwidth. To allow for long distance transmission, the gain specification goal is 60dB of voltage gain within $\pm 1.5\text{dB}$ over the full bandwidth of at 500kHz – 7.2MHz. The maximum allocated dc current target goal is 12mA or less from a voltage supply of +9V.

Circuit Requirements

- $V_o/V_i = 60 \pm 1.5\text{ dB}$, which is 1000 V/V. (High)
- Broadband width is from 0.5 to 7.2 MHz.
- $R_{in} = 1.5\text{ k}\Omega \pm 20\%$ (Moderate)
- $R_{out} = 75\text{ }\Omega \pm 20\%$ (Low)
- Power = $\leq 12\text{mA}$ from +9V supply
- Survive $\pm 50\% \beta_o$ Variations

It can be noted that the input and output specifications exist to impedance match the amplifier with the SA612 which features a $1.5\text{ k}\Omega$ output impedance and the 75Ω antenna.

There is a penalty for utilizing over 60mW of power of \$0.15 per mW above that threshold.

1.1 Transistor Definitions

Transistor Definitions

We used two transistors throughout the entire amplifier: the **ZTX107** from Zetex and the **DSC5F01** from Panasonic.

Our calculations are based on the following assumptions:

ZTX107

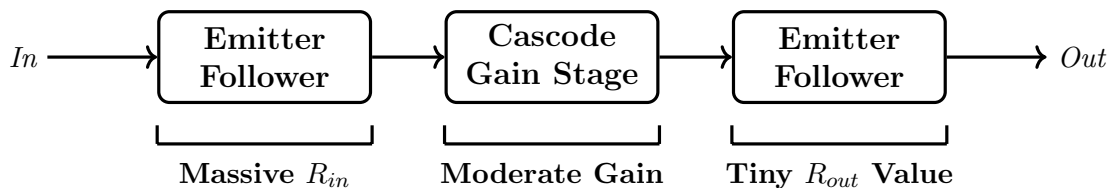
- $H_{fe} = 250\text{--}300$
- $f_t = 110 - 180 \text{ MHz}$
- $C_\mu = 4\text{--}5 \text{ pF}$
- $C_\pi = 30\text{--}50 \text{ pF}$

DSC5F01

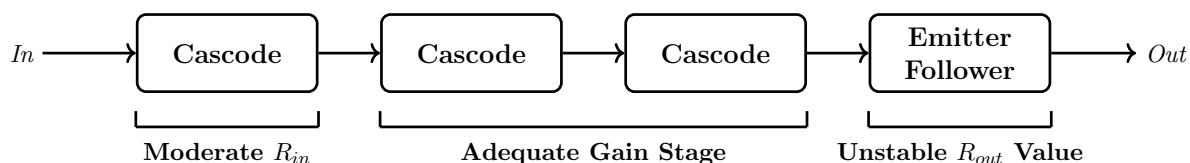
- $H_{fe} = 40\text{--}100$
- $f_t = 1.9 \text{ GHz}$
- $C_\mu = 1\text{--}2 \text{ pF}$
- $C_\pi = 0.6\text{--}2 \text{ pF}$

2 Conceptual Design

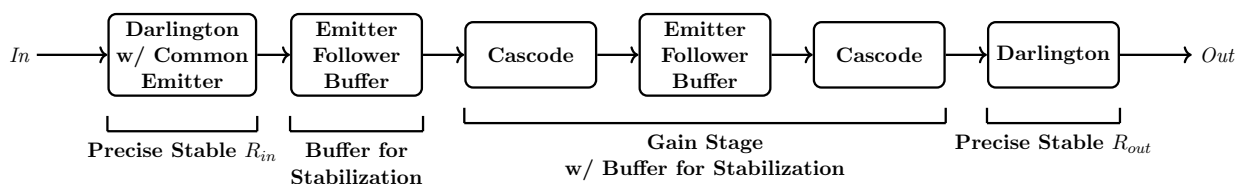
Original Design



Second Design



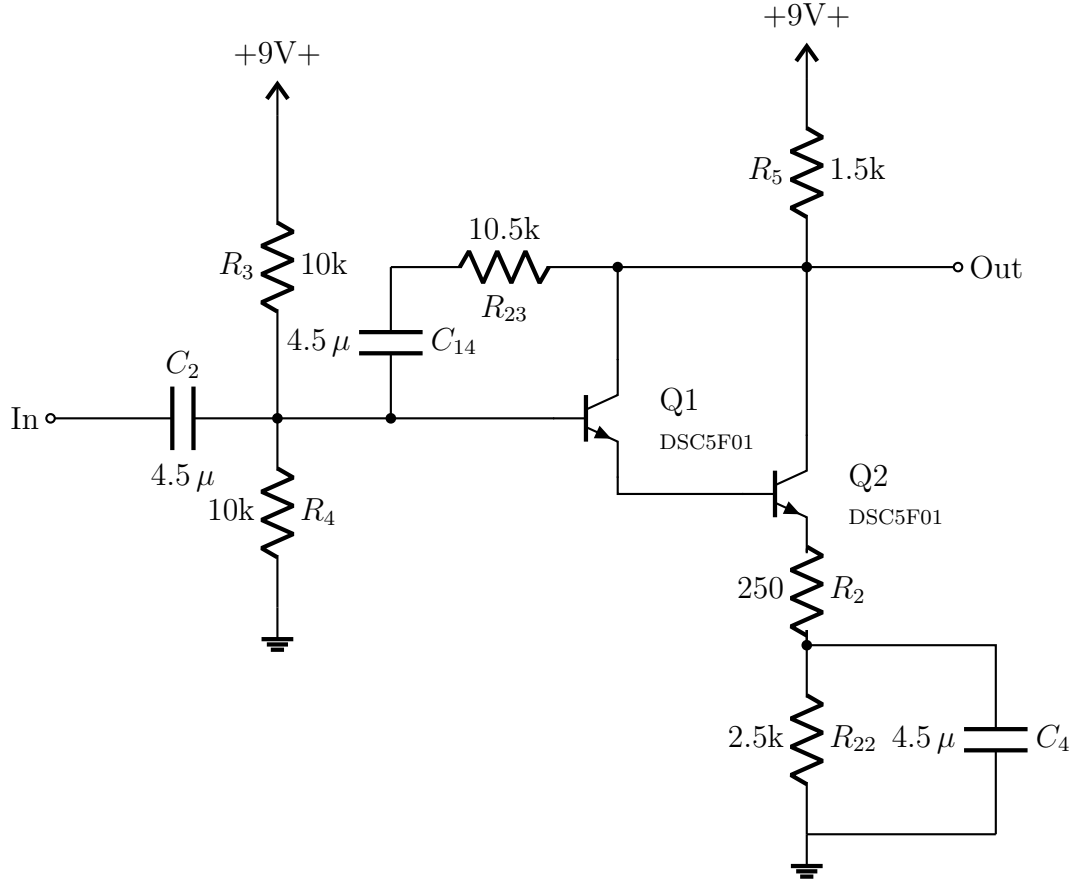
Final Design



Our initial amplifier design started with an emitter follower, followed by a single cascode, and then another emitter follower. This design had multiple problems including frequency response, too large of an R_{in} value, and too little gain. Our second design features a cascode for the input which handled the R_{in} specification better than the previous due to its moderate input resistance but still struggled with variations in β_o preventing our design from hitting the target $1.5 \text{ k}\Omega \pm 20\%$. We also switched to two cascode amplifier for our gain stage which proved adequate for our target of 60 dB, but suffered from poor frequency response. We continued to use an emitter follower for our output which we were able to tune to a 75Ω output but it was too variable. In our improved final design we introduced several new elements starting with a darlington amplifier that features a common emitter as the second transistor. Utilizing feedback, we were able to achieve a very stable R_{in} value that meets the specification. To protect the poles and frequency response of our first cascode in the gain stage we introduced an emitter follower buffer. We also repeated this process inside of our gain stage as it proved very effective. In addition to the buffer inside the gain stage we used feedback to stabilize our gain in the face of varying β_o values. We ended our design with a darlington amplifier that featured two emitter followers and produced a very stable 75Ω output well within the specification of $\pm 20\%$.

3 Design Implementation

3.1 Input Stage

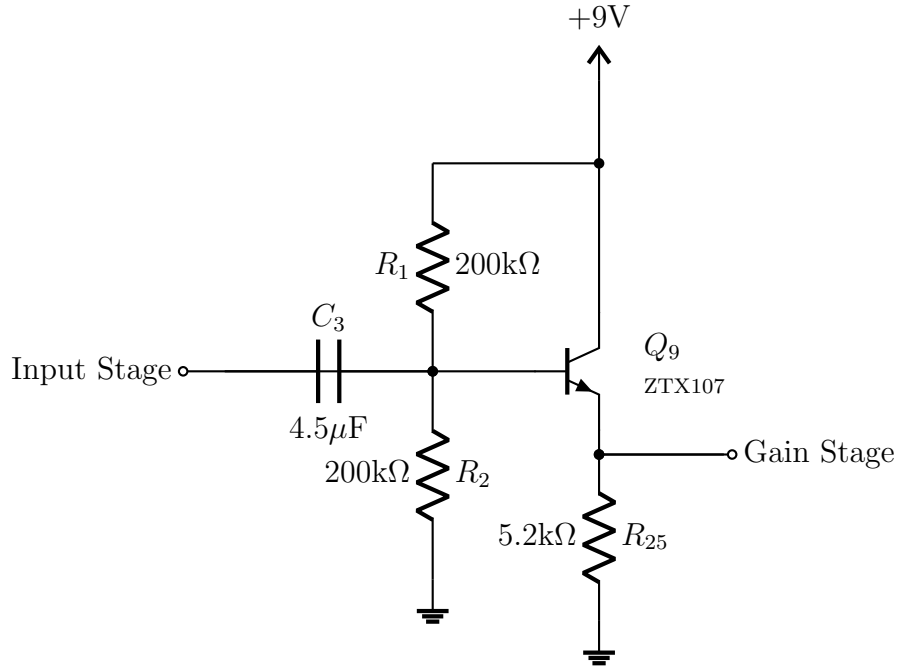


Parameter	Symbol	Calculated Value
Base Voltage 1	V_{B1}	4.499 V
Base Voltage 2	V_{B2}	3.799 V
Emitter Voltage 2	V_{E2}	3.099 V
Collector Voltage	V_C	7.327 V
Emitter Current 2	I_{E2}	1.127 mA
Base Current 2	I_{B2}	11.737 μ A
Collector Current 2	I_{C2}	1.115 mA
Emitter Current 1	I_{E1}	11.737 μ A
Base Current 1	I_{B1}	286.27 nA
Collector Current 1	I_{C1}	11.157 μ A
Input Resistance	R_{IN}	1.3 k Ω
Voltage Gain	A_v	-5.049 V/V
Cutoff Frequency	F_H	9.98 MHz

1. Assumed values for this analysis: $\beta_1 = 40$, $\beta_2 = 95$.

Table 1: DC Operating Point Results

3.2 Emitter Follower Buffer 1

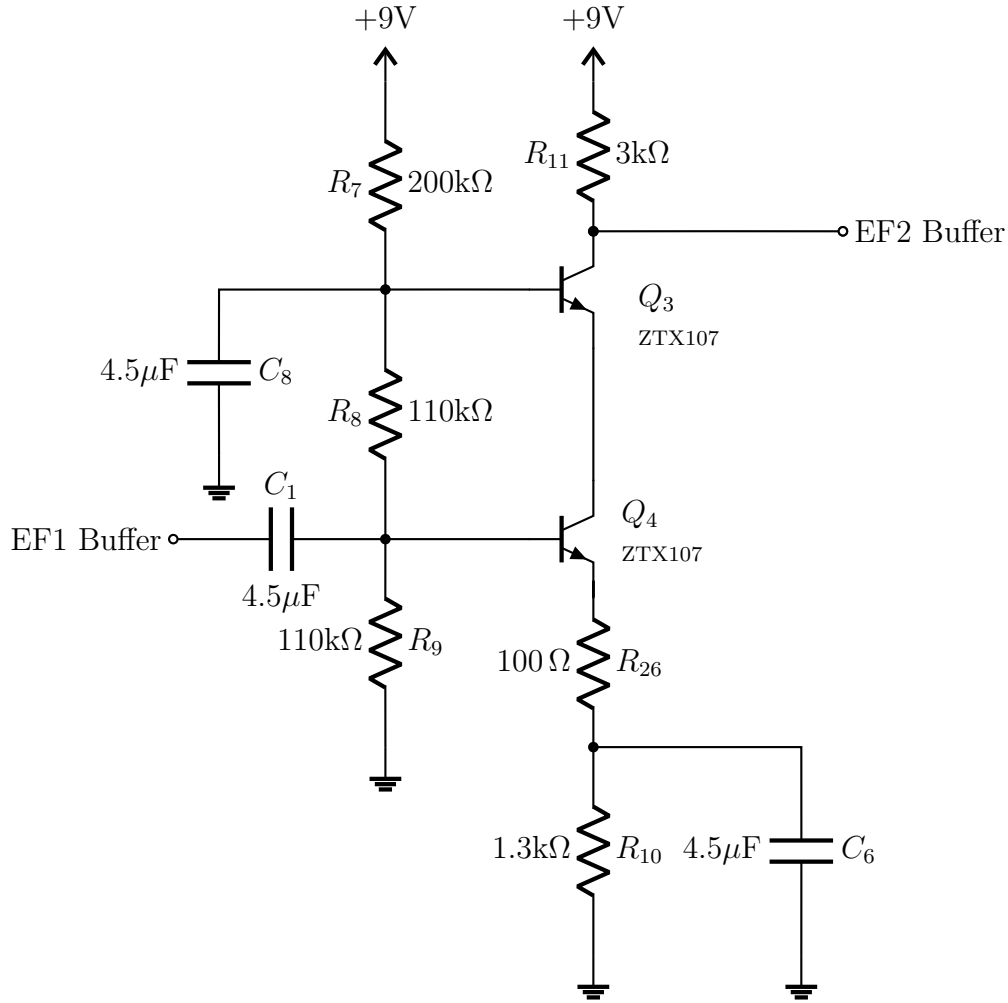


Parameter	Symbol	Calculated Value
Emitter Current	i_e	0.7028 mA
Base Current	i_b	2.8 μ A
Emitter Voltage	V_E	3.6 V
Base Voltage	V_{B2}	$\approx$ 4.26 V
Transconductance	g_m	0.0269 A/V
Transistor Input Resistance	r_π	9290 Ω
Stage Input Resistance	R_{in}	91k Ω
Bias Resistor 1	R_1	200k Ω
Bias Resistor 2	R_2	200k Ω
Voltage Gain	A_v	-0.1 dB
Cutoff Frequency	F_H	30.7 MHz

1. Assumed values for this analysis: $\beta = 250$, $I_C = 0.7$ mA.

Table 2: Summary of DC Operating Point and Small Signal Parameters

3.3 Gain Cascode Stage 1

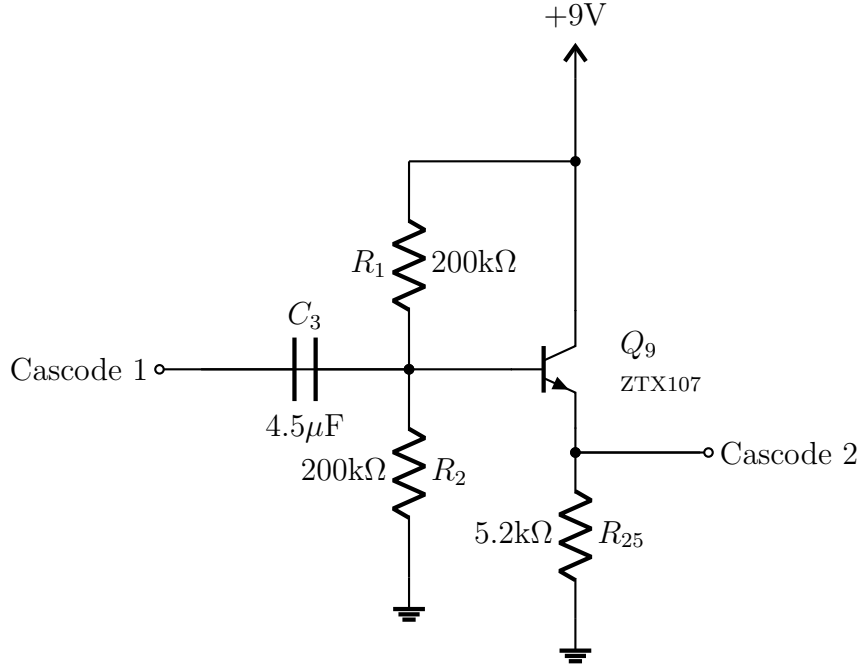


Parameter	Symbol	Calculated Value
Base Current	i_b	$3 \mu\text{A}$
Emitter Voltage	V_E	1.26 V
Lower Base Voltage	V_{B1}	$\approx 1.85 \text{ V}$
Upper Base Voltage	V_{B2}	$\approx 4.23 \text{ V}$
Transconductance	g_m	0.0346 A/V
Transistor Input Resistance	r_π	8600Ω
Stage Input Resistance	R_{in}	22000Ω
Bias Resistor 1	R_1	$200\text{k} \Omega$
Bias Resistor 2	R_2	$110\text{k} \Omega$
Bias Resistor 3	R_3	$110\text{k} \Omega$
Voltage Gain	A_v	27 dB
Cutoff Frequency	F_H	13.6MHz

1. Assumed values for this analysis: $\beta = 300$, $I_C = 1.15 \text{ mA}$.

Table 3: Summary of DC Operating Point and Small Signal Parameters

3.4 Emitter follower Buffer 2

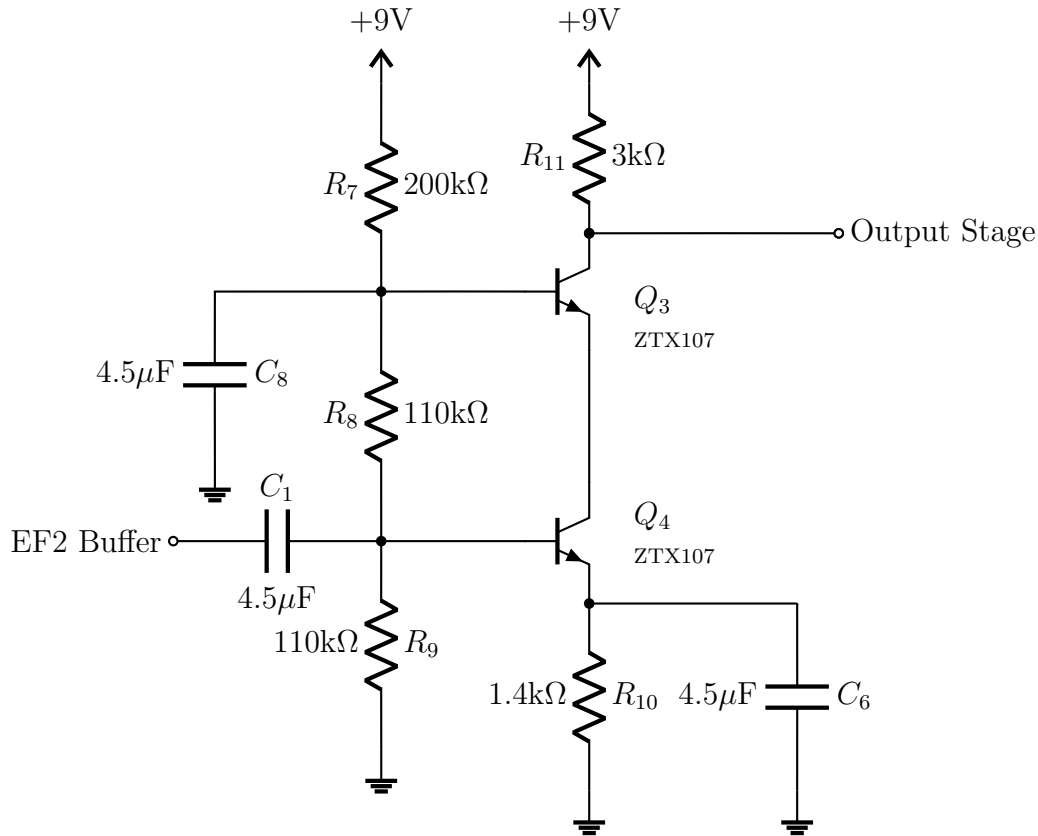


Parameter	Symbol	Calculated Value
Base Current	i_b	$2.8 \mu\text{A}$
Emitter Voltage	V_E	3.6 V
Base Voltage	V_{B2}	$\approx 4.26 \text{ V}$
Transconductance	g_m	0.0269 A/V
Transistor Input Resistance	r_π	9290Ω
Stage Input Resistance	R_{in}	$88\text{k} \Omega$
Bias Resistor 1	R_1	$200\text{k} \Omega$
Bias Resistor 2	R_2	$200\text{k} \Omega$
Voltage Gain	Av	-0.1 dB
Cutoff Frequency	F_H	13.7MHz

1. Assumed values for this analysis: $\beta = 250$, $I_C = 0.7 \text{ mA}$.

Table 4: Summary of DC Operating Point and Small Signal Parameters

3.5 Gain Cascode Stage 2

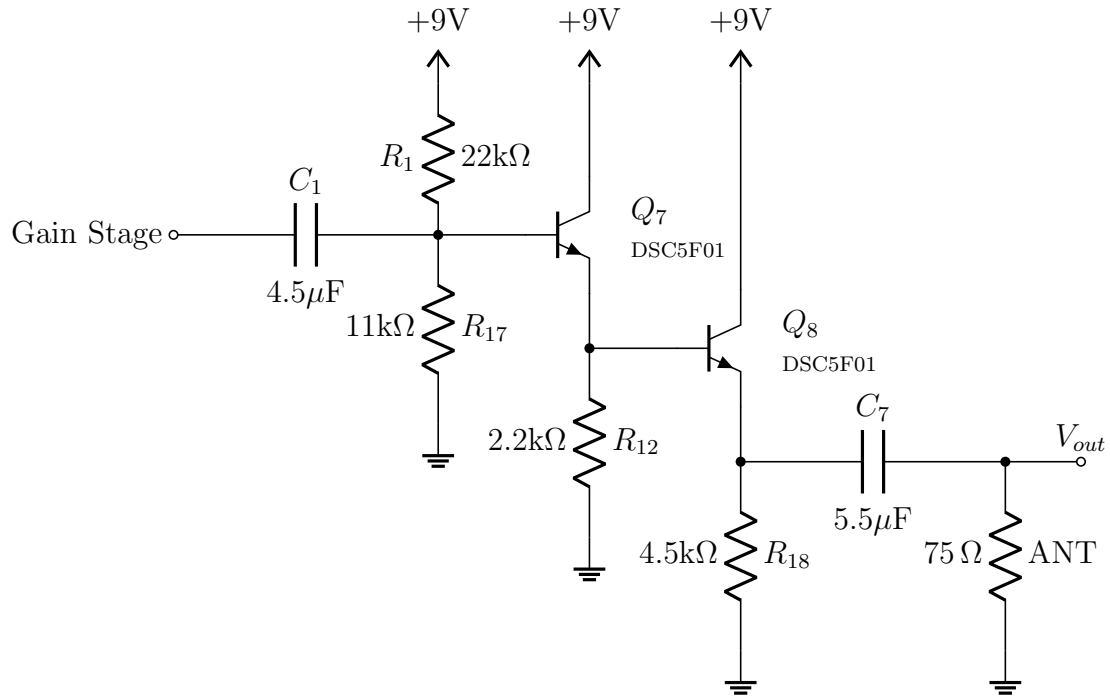


Parameter	Symbol	Calculated Value
Base Current	i_b	$3 \mu\text{A}$
Emitter Voltage	V_E	1.26 V
Lower Base Voltage	V_{B1}	$\approx 1.85 \text{ V}$
Upper Base Voltage	V_{B2}	$\approx 4.23 \text{ V}$
Transconductance	g_m	0.0346 A/V
Transistor Input Resistance	r_π	8600Ω
Stage Input Resistance	R_{in}	7500Ω
Bias Resistor 1	R_1	$200\text{k} \Omega$
Bias Resistor 2	R_2	$110\text{k} \Omega$
Bias Resistor 3	R_3	$110\text{k} \Omega$
Voltage Gain	A_v	37.2 dB
Cutoff Frequency	F_H	24.5MHz

1. Assumed values for this analysis: $\beta = 300$, $I_C = 1.15 \text{ mA}$.

Table 5: Summary of DC Operating Point and Small Signal Parameters

3.6 Darlington Output Stage



Parameter	Symbol	Calculated Value
Base Voltage 1	V_{B1}	2.918 V
Base Voltage 2	V_{B2}	2.218 V
Emitter Voltage 1	V_{E1}	2.218 V
Emitter Voltage 2	V_{E2}	1.518 V
Emitter Current 2	I_{E2}	0.337 mA
Base Current 2	I_{B2}	3.708 μ A
Collector Current 2	I_{C2}	0.334 mA
Emitter Current 1	I_{E1}	1.012 mA
Base Current 1	I_{B1}	11.122 μ A
Collector Current 1	I_{C1}	1.001 mA
Voltage Gain	A_v	-6.2 dB
Cutoff Frequency	F_H	15.3 MHz

1. Assumed values for this analysis: $\beta_1 = 90$, $\beta_2 = 90$.

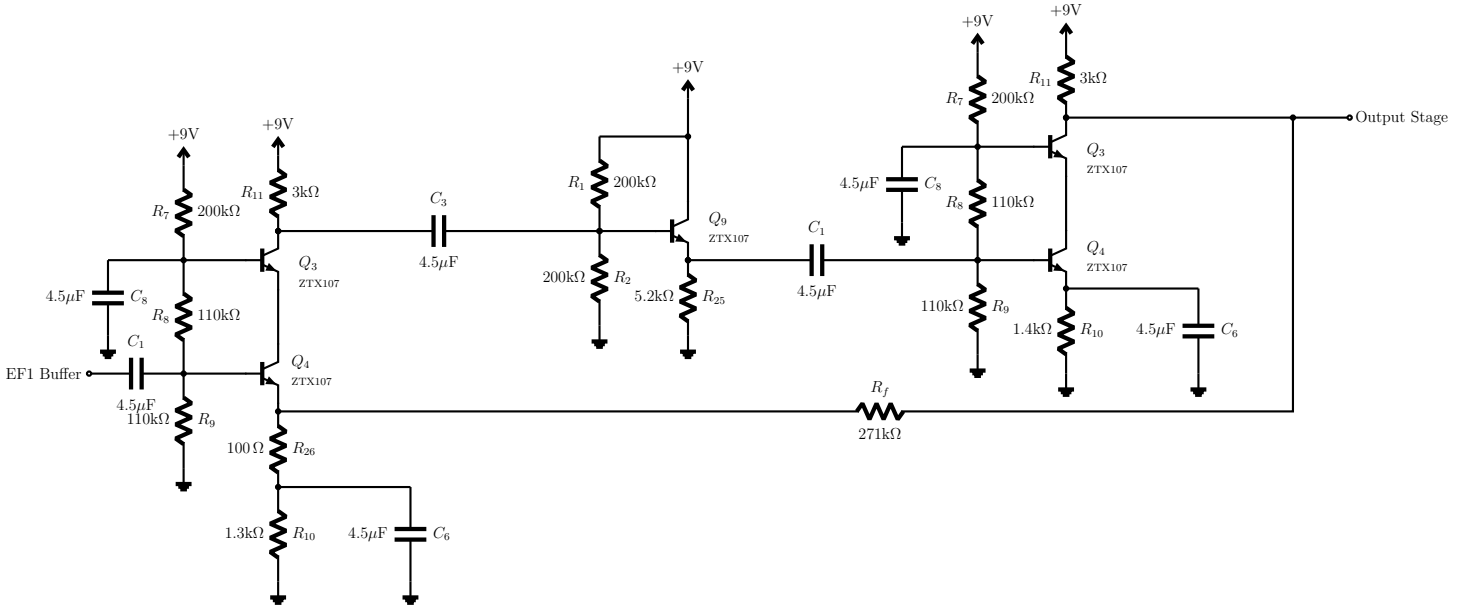
Table 6: DC Operating Point Results

3.7 Feedback

We implemented feedback in two major sections of our design: the input and the gain stages. We decided to use feedback to reduce and stabilize the input impedance of the input stage to meet our specifications but ran into an issue with the typical darlington amplifier which consist of an EF-EF style cascade. We decided to implement shunt-shunt feedback which involves sampling the voltage at the output of the amplifier and using this to control the input. With the EF-EF style of darlington, we were not seeing a high enough voltage gain for feedback to be properly implemented so we decided to switch the second transistor to a common emitter configuration to boost our voltage gain. This boost in voltage gain created the perfect conditions for us to manipulate the millers impedance of the feedback resistor to achieve our target R_{in} value. This process is described in here

Our main Gain stage consists of a cascode, emitter follower, and another cascode. Prior to its current implementation the entire gain stage exhibited 77dB of gain. Additionally, the gain was extremely variable and susceptible to changes in variations across β_o . To account for this we introduced series/shunt feedback to reduce our gain and also stabilize it within the specified bandwidth.

The feedback implemented gain stage is shown below.



The series/shunt feedback was connected from the emitter of the first cascode to the collector of the second cascode, and the R_F resistor value was 271kΩ. Our feedback analysis results yielded a new midband gain of 60dB through approximately 12 MHz. Our Simulation results concurred showcasing a nominal feedback gain of 59.74 dB in the passband of interest with a maximum and minimum of 60.4dB and 58.7dB respectively. The indepth feedback analysis can be seen on page 33.

4 Simulation Experiment

The simulation component of our design development was critical to design assessment and adjustment. Aside from not having immediate access to all design components, simulations are much more cost effective than physical testing.

We utilized LTspice for all of our simulations. The transistor models were provided by the instructor. We used the AC analysis simulation type for all tests. Within the provided transistor models were 3 categories, low, nominal, and high, that provided different β_o values for the transistors. In every single simulation we used one +9V source to maintain the ease in tracking total power consumption.

4.0.1 Simulation Circuit Setup

To test our design we started by simulating each stage separately. We designed the circuit starting with transistors in their nominal condition.

4.0.2 DC Characteristics and Assumptions

Our initial step when testing each was to run the .op command in Ltspice to verify our DC calculations. We would also use ".option logopinfo" to verify our parasitic capacitance and β_o assumptions.

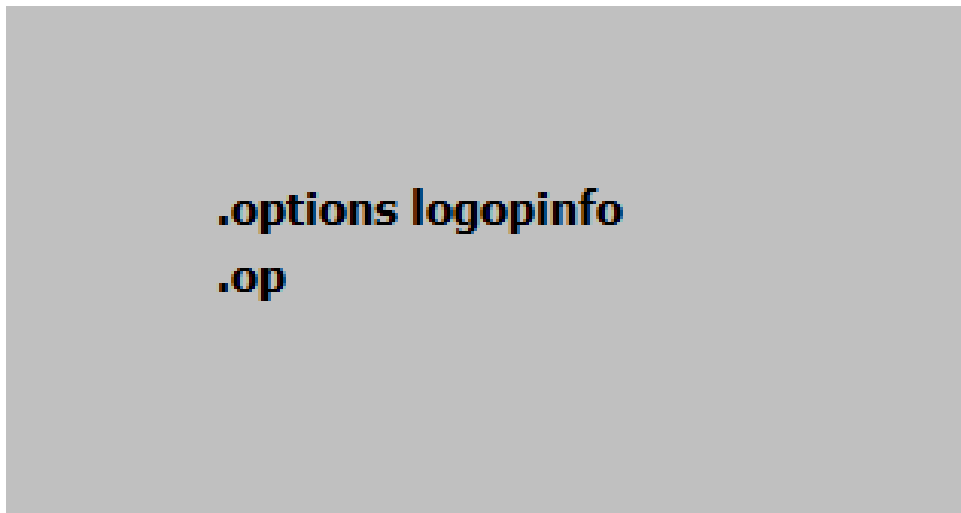


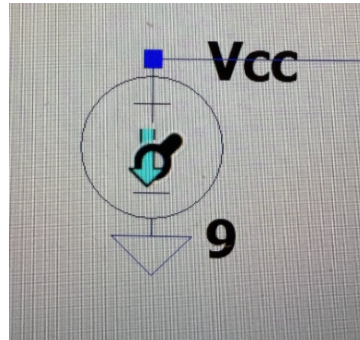
Figure 1: DC operation point simulation.

4.0.3 AC Characteristics

Following the verification of transistor states and DC characteristics we assess the critical AC characteristics of each stage. For the input stage this was the value of the input impedance, the gain for the middle stages, and the output impedance for our output stage.

To perform resistance calculations, we configured an AC voltage source (AC amplitude of 1 all other settings blank) with its positive pole oriented towards the decoupling capacitor associated with the node of interest. We plotted this voltage divided by the current traveling through the decoupling capacitor to derive AC resistance seen in the node.

For power and current simulation, AC analysis was on the full schematic and mouse was hovered over the power source to read the values.



DC operating point: I(V3) = -6.5079276mA Dissipation=-58.571348mW

Figure 2: Power and current draw reading

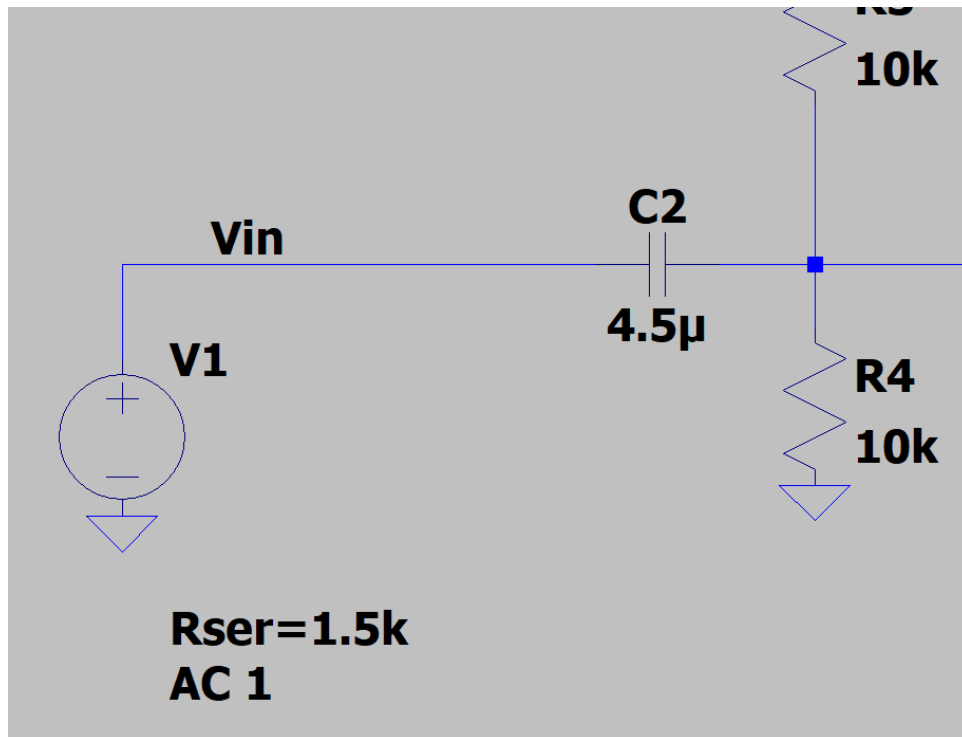


Figure 3: For the R_{IN} simulation, we plotted V_{IN} over the current going through the C_2 capacitor.

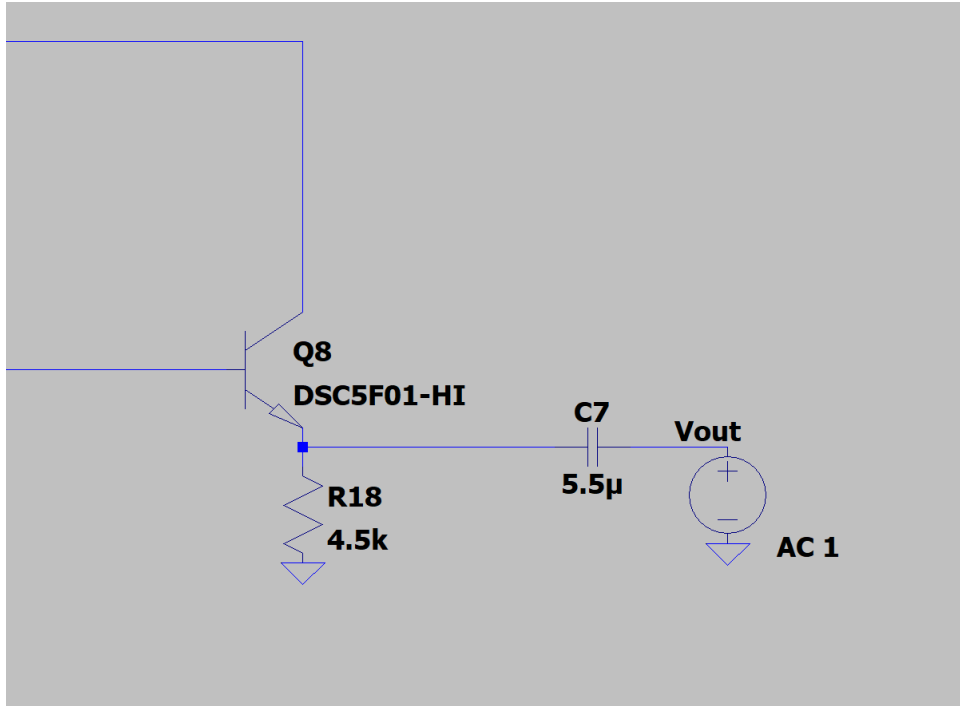


Figure 4: For the R_{OUT} simulation, we plotted V_{OUT} over the current going through the C_7 capacitor.

To perform Midband gain calculations we configured an AC voltage source to have an AC amplitude of 1, and AC series resistance of whatever the previous stage possessed. In the case of the input stage this would be 1.5 k Ω for the SA612A, and in following stages this would be the output impedance of the previous stage. We also simulated loads of subsequent loads of next stages using a resistor between the output on the stage of focus and ground that was of the approximate value of R_{in} for the next stage. We then plotted the voltage at the output node of that specific stage. To plot the gain over the frequency, the AC analysis was run and V_{IN} over V_{OUT} trace was added.

4.0.4 Complete Circuit

For complete circuit testing the procedures repeated for the individual stages to find important resistances and midband gain were repeated but on for the entire circuit composed together. It was common to have issues arise when the circuit was combined into one, and so a useful tactic that we utilized was plotting specific voltages and monitoring gain or current over portions of the complete circuit instead of the total output. This allowed us to better understand the loading effects and interaction of our stages to make necessary adjustments.

5 Results / Analysis

5.0.1 LO

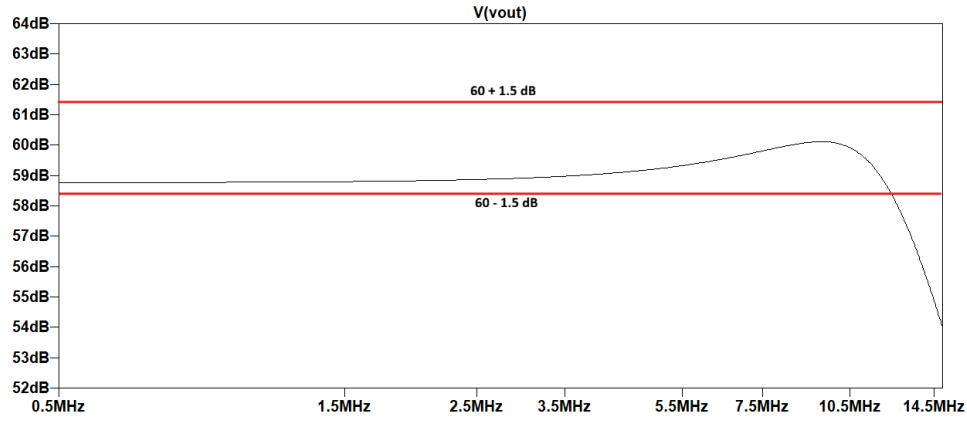


Figure 5: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

As it shown in the picture, our design meets the gain requirements. It maintains the gain between 58.5dB and 61.5dB within the required bandwidth of 0.5-7.1MHz. Apart from this, the cutoff frequency is 14MHz, which demonstrates that our circuit met the bandwidth spec.

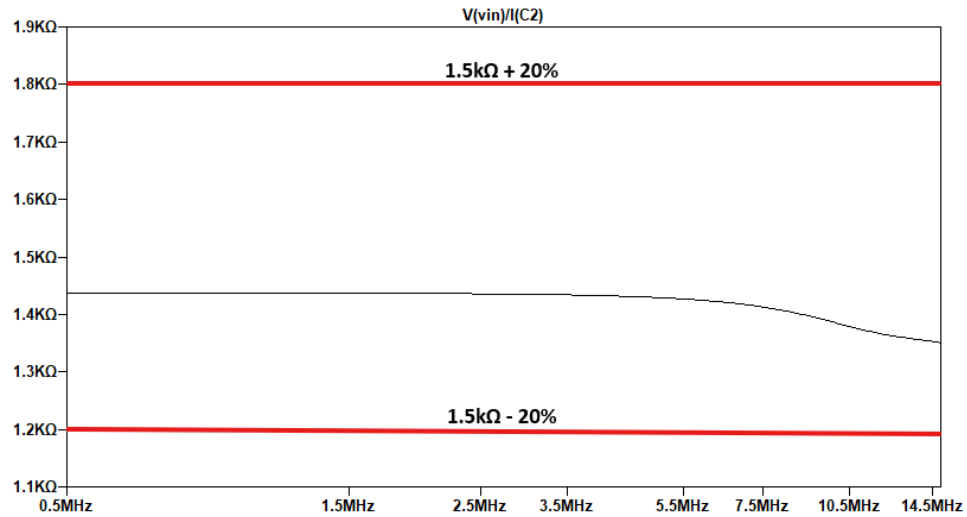


Figure 6: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

This plot illustrates that our design meets the R_{IN} requirement, as the input resistance is within $1.5 \text{ k}\Omega \pm 20\%$.

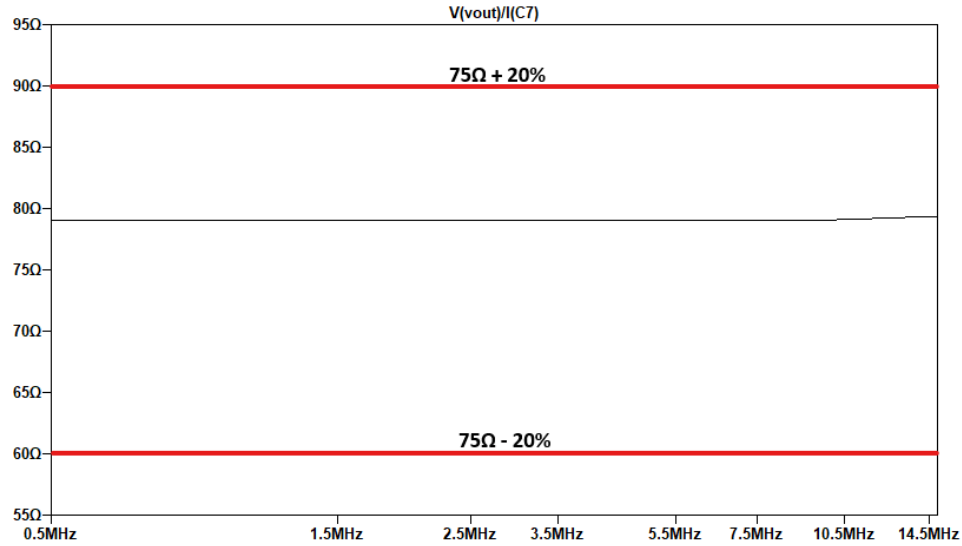


Figure 7: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

This plot illustrates that our design successfully meets the R_{OUT} requirement, as the output resistance is within $75\Omega \pm 20\%$.

5.0.2 NOM

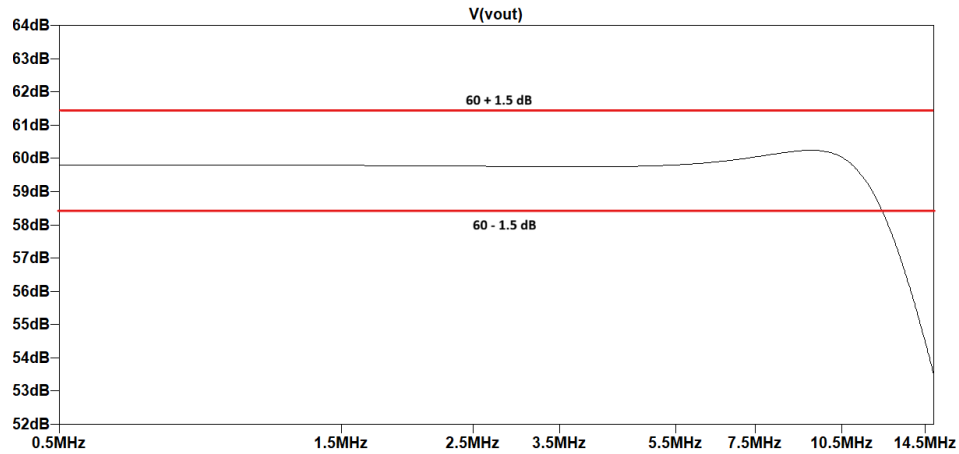


Figure 8: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

As it shown in the picture, our design meets the gain requirements. It maintains the gain between 58.5dB and 61.5dB within the required bandwidth of 0.5-7.1MHz. Apart from this, the cutoff frequency is 14MHz, which demonstrates that our circuit met the bandwidth spec.

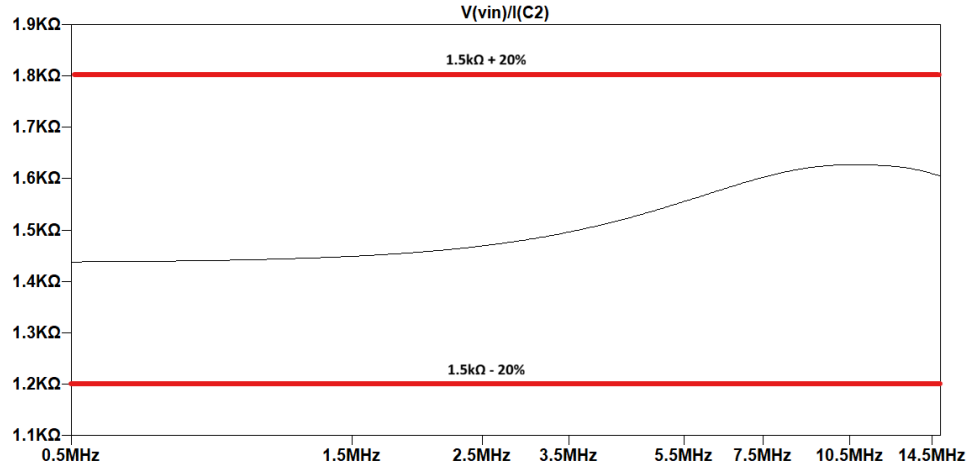


Figure 9: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

This plot illustrates that our design meets the R_{IN} requirement, as the input resistance is within $1.5 \text{ k}\Omega \pm 20\%$.

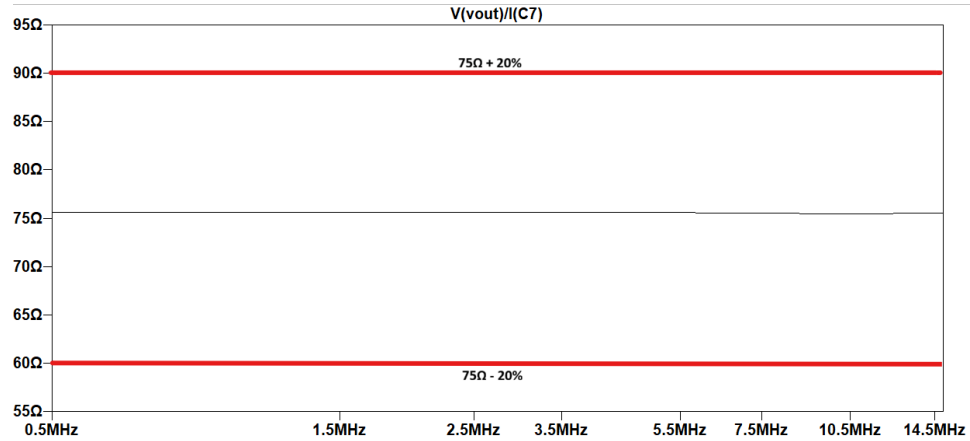


Figure 10: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

This plot illustrates that our design successfully meets the R_{OUT} requirement, as the output resistance is within $75\Omega \pm 20\%$.

5.0.3 HI

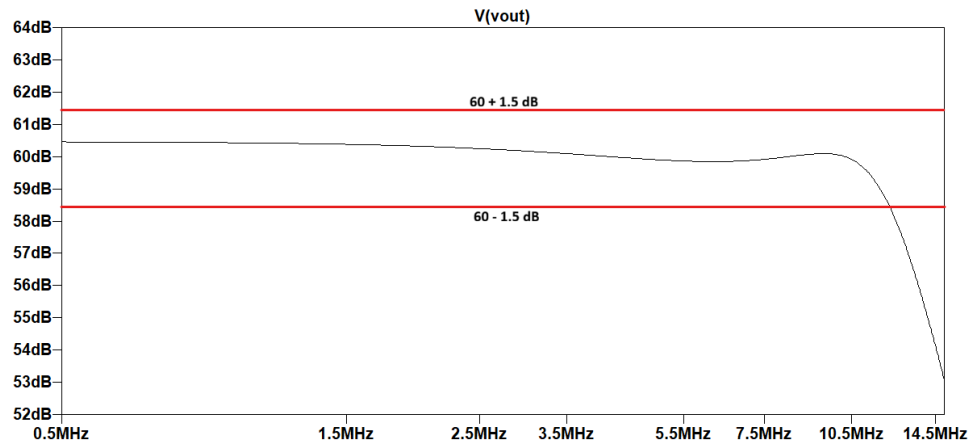


Figure 11: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

As it shown in the picture, our design meets the gain requirements. It maintains the gain between 58.5dB and 61.5dB within the required bandwidth of 0.5-7.1MHz. Apart from this, the cutoff frequency is 14MHz, which demonstrates that our circuit met the bandwidth spec.

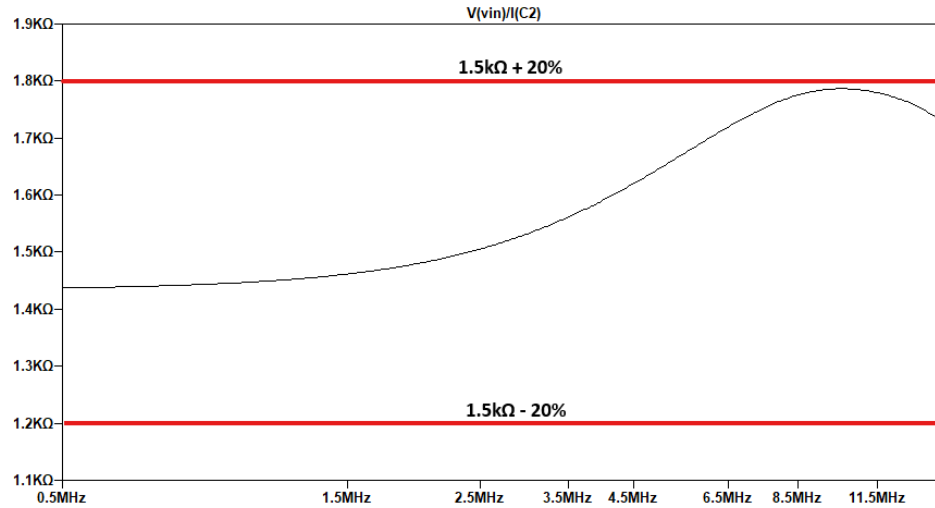


Figure 12: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

This plot illustrates that our design meets the R_{IN} requirement, as the input resistance is within $1.5 \text{ k}\Omega \pm 20\%$.

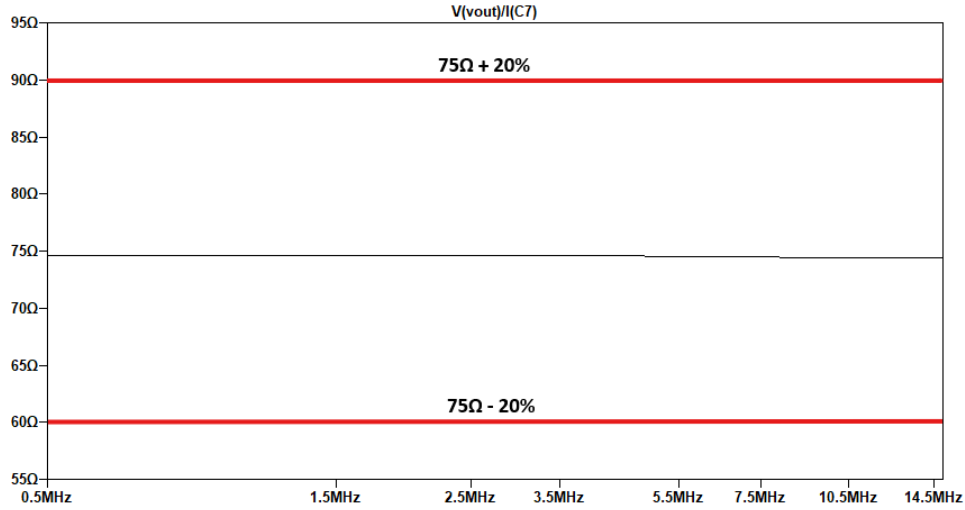


Figure 13: The gain of the circuit over the frequency range between 0.5MHz to 15MHz

Discussion

This plot illustrates that our design successfully meets the R_{OUT} requirement, as the output resistance is within $75\Omega \pm 20\%$.

5.0.4 Power and Current Draw

	Low	Nominal	High
Power	56.2 mW	58.6 mW	60.4 mW

Discussion

Our design successfully met the power requirements across all tested operating conditions, aligning with the 60 mW target specification.

	Low	Nominal	High
Current draw	6.2 mA	6.5 mA	6.7 mA

Discussion

Our design successfully met the current draw requirements across all tested operating conditions, aligning with under 12 mA.

6 Conclusions

Designing an RF amplifier with strict gain, bandwidth, and impedance requirements underscored the necessity of a modular, iterative workflow. Two significant lessons learned from this project were the importance of a "divide and conquer" methodology and the ability to recognize and abandon "dead-end" designs. While pivoting after significant labor is difficult, we found that early identification of a topology's inherent limitations is far more efficient than attempting to force a sub-optimal design to meet stringent specifications. This project also highlighted that high-level cooperation and communication are essential to producing a technical report of this caliber.

The selection of amplifier topologies highlighted the critical trade-offs between gain and bandwidth. While the common-emitter configuration provided the necessary gain, its inherent frequency response limitations (due to the Miller effect) necessitated the strategic use of emitter followers as buffers to maintain stage isolation. Furthermore, the integration of Darlington pairs proved vital for achieving high input/output impedances; by leveraging the compounded β , we met specifications that standard single-transistor configurations could not support.

A significant challenge in the early design phase was reconciling hand-calculated parameters with simulation models. Parameters such as base-emitter capacitance (C_π), base-collector capacitance (C_μ), and β were initially pulled from datasheets. However, LTspice simulations revealed β variations that deviated from our ideal assumptions. We successfully mitigated this by utilizing the Operating Point (.op) analysis feature to verify and tune our hand calculations, ensuring our theoretical model aligned with the simulated transistor physics.

Ultimately, the most significant technical takeaway was the nuanced role of negative feedback. While effective for extending bandwidth and enhancing stability, we concluded that feedback is not a "fix" for a poor open-loop design. Our observation of gain peaking—the "hump" in the frequency response—demonstrated that pushing feedback limits on an uncharacterized circuit compromises the phase margin. Successful RF design, therefore, relies on a well-stabilized open-loop foundation before feedback is applied to fine-tune performance.

7 Appendices

7.1 Appendix A / Equations

DC

$$V_{BE} = 0.7 \text{ V} \quad (\text{nnp})$$

$$V_{EB} = 0.7 \text{ V} \quad (\text{pnp})$$

$$i_C = \beta i_B$$

$$i_C = \alpha i_E \quad \text{or} \quad i_E = (\beta + 1)i_B$$

$$V_{CE} = V_{CB} + V_{BE} \quad (\text{nnp})$$

$$V_{EC} = V_{EB} + V_{BC} \quad (\text{pnp})$$

Small Signal

$$g_m = I_{CQ}/V_t$$

$$\beta_o = g_m r_\pi$$

$$r_o = V_A/I_{CQ}$$

$$\omega_T = \frac{\beta_o}{r_\pi(C_\mu + C_\pi)} = \frac{g_m}{(C_\mu + C_\pi)}$$

BJT AMPLIFIER CONFIGURATIONS

Common Emitter:

$$A_v = -g_m(R_C \parallel R_L)$$

$$R_{in} = R_{BIAS} \parallel r_\pi$$

$$R_{out} = r_o \parallel R_C$$

Emitter Follower:

$$A_v = \frac{(1 + \beta_o)(R_E \parallel R_L)}{r_\pi + (1 + \beta_o)(R_E \parallel R_L)}$$

$$R_{in} = R_{BIAS} \parallel [r_\pi + (1 + \beta_o)(R_E \parallel R_L)]$$

$$R_{out} = R_E \parallel \frac{r_\pi + R_S \parallel R_{BIAS}}{(1 + \beta_o)}$$

Common Emitter with emitter resistor:

$$A_v = \frac{-\beta_o(R_C \parallel R_L)}{r_\pi + (1 + \beta_o)R_E}$$

$$R_{in} = R_{BIAS} \parallel [r_\pi + (1 + \beta_o)R_E]$$

$$R_{out} \cong R_C$$

Common Base:

$$\begin{aligned}A_v &\cong g_m(R_C \parallel R_L) \\ R_{in} &\cong 1/g_m \\ R_{out} &\cong R_C\end{aligned}$$

Cascode

$$A_{vo} = -\frac{R_{in}}{R_S + R_{in}} \cdot g_m(R_C \parallel R_L) \quad R_{in} = r_\pi \parallel R_2 \parallel R_3$$

Input pole due to C_{in} :

$$\begin{aligned}C_{in} &= C_{\pi 1} + 2C_\mu && \leftarrow \text{small} \\ R_{eq} &= r_{\pi 1} \parallel R_2 \parallel R_3 \parallel R_S && \leftarrow \text{moderate} \\ f_1 &= \frac{1}{2\pi C_{in} R_{eq}} && \leftarrow \text{large (hi freq)}\end{aligned}$$

Intermediate pole due to C_m :

$$\begin{aligned}C_m &= C_{\pi 2} + C_{\mu 1} && \leftarrow \text{small} \\ R_{eq} &= 1/g_{m2} && \leftarrow \text{very small} \\ f_2 &= \frac{1}{2\pi C_m R_{eq}} && \leftarrow \text{very large (very hi freq, can ignore)}\end{aligned}$$

Output pole due to C_{out} :

$$\begin{aligned}C_{out} &= C_{\mu 2} && \leftarrow \text{small} \\ R_{eq} &= R_C \parallel R_L && \leftarrow \text{moderate} \\ f_3 &= \frac{1}{2\pi C_{out} R_{eq}} && \leftarrow \text{large (hi freq)}\end{aligned}$$

7.2 Appendix B / Hand Calculations

7.2.1 Input Stage

DC Characteristics

The darlington amplifier configuration was chosen due to its high current gain and more importantly high input impedance. For the gain of our common emitter, which is talked about in a later section we wanted to provide an adequate amount of current, around 1.15 mA. Our values were designed around that goal and we used the relationships defined in the table below. Our hand-calculated values using these relationships came out to the numbers displayed in the results table.

Rin & Midband Gain

$$\begin{aligned}
 v_i &= r_{\pi 1} i_{b1} + r_{\pi 2} i_{b2} + (i_{b2} + \beta_2 i_{b2}) R_E \\
 &= r_{\pi 1} i_{b1} + r_{\pi 2} (\beta_1 + 1) i_{b1} + \left((\beta_1 + 1) i_{b1} + \beta_2 ((\beta_1 + 1) i_{b1}) \right) R_E \\
 &= i_b \left[r_{\pi 1} + r_{\pi 2} (\beta_1 + 1) + [\beta_1 + 1 + \beta_2 (\beta_1 + 1)] R_E \right] \\
 &= i_b \left\{ r_{\pi 1} + (\beta_1 + 1) [r_{\pi 2} + (1 + \beta_2) R_E] \right\}
 \end{aligned}$$

$$\begin{aligned}
 v_o &= -(\beta_1 i_{b1} + \beta_2 i_{b2}) R'_L \\
 &= -i_b (\beta_1 + \beta_2 (\beta_1 + 1)) R'_L \\
 \frac{v_o}{v_i} &= \frac{-[\beta_1 + \beta_2 (\beta_1 + 1)] R'_L}{r_{\pi 1} + (\beta_1 + 1) [r_{\pi 2} + (1 + \beta_2) R_E]}
 \end{aligned}$$

We decided to use feedback to reduce and stabilize the input impedance to meet our parameters but ran into an issue with the typical darlington amplifier which consist of an EF-EF style cascade. We decided to implement shunt-shunt feedback which involves sampling the voltage at the output of the amplifier and using this to control the input. With the EF-EF style of darlington, we were not seeing a high enough voltage gain for feedback to be properly implemented so we decided to switch the second transistor to a common emitter to boost our voltage gain. This boost in voltage gain created the perfect conditions for us to manipulate the millers impedance to achieve our R_{in} value. We derived the small signal open loop voltage gain of amplifier using the small signal model and derivation depicted above. Filling in for the values, we arrive at a A_v of -5.049.

This is critical because with our feedback resistor the R_{in} is defined as

$$R_{in} = 5k\Omega \parallel Z_1 \parallel \left[r_{\pi 1} + (\beta_1 + 1)(r_{\pi 2} + \beta_2(250)) \right]$$

Millers theorem tells us that

$$Z_1 = \frac{Z}{1 - A_v}$$

this corresponds to a Z_1 of 1750Ω which when paired with the $5 k\Omega$ yields an approximate Rin of $1.3k\Omega$. This input varies from roughly 1250 to 1770Ω depending on the beta values

assumed during previous calculation but confidently stays within our specification target.

Frequency Response

There are five poles in our input section, however four of those poles are interactive on their, and the last pole is interactive with other poles in our trailing stages. To account for this we calculated the time constants of c_{m2} and c_{m2} , and the complete pole for the other capacitances of interest.

The capacitances on both transistors were split into millers capacitances. Starting with c_{m1} , the R_{eq} is approximately R_1 in parallel with R_{in1} . R_1 dominates as it is the parallel combination: $1.5\text{ k}\Omega \parallel 5\text{ k}\Omega \parallel 1.3\text{ k}\Omega$ from feedback. This total comes out to approximately 611Ω s. The c_{eq} is based on our closed loop gain of approximately 5 producing a $c_{eq} = C_\mu(1 - A_v) = 6\text{ pF}$. This leads to a time constant of $\tau_1 = 3.666 \times 10^{-9}$.

Our next capacitance $c_{\pi1}$ is approximately 0.6 pF . Using the standard equation for the emitter follower resistance, this poles R_{eq} comes out to $2.1\text{ k}\Omega$. This leads to a time constant of $\tau_2 = 1.285 \times 10^{-9}$.

The next pole containing c_{m3} is the most dominant pole. The gain in determining this value is based on the CE element gain which due to our emitter resistor is only -5.45. This leads to a c_{eq} of 6.45 pF . The R_{eq} is the resistance looking into the emitter which using the standard equation is approximately $1.3\text{ k}\Omega$ and the resistance looking into base of transistor 2 using the standard equation is approximately $27.5\text{ k}\Omega$. The total R_{eq} is $1.2\text{ k}\Omega$. This yields $\tau_3 = 7.39 \times 10^{-9}$.

The next time constant is $c_{\pi2}$ which follows a similar equation to $c_{\pi1}$ yielding an R_{eq} of 205Ω and $\tau_4 = 3.6 \times 10^{-9}$.

Summing all of those together we get a pole response of 9.98 MHz .

We have one non interactive pole that is a combination of c_{m2} and c_{m2} which utilizes millers leads to a combined capacitance of 2.4 pF . This combines with the c_μ on the emitter follower buffer leading to a total equivalent capacitance of 6.4 pF and the R_{eq} can be visually assessed at $1.5\text{ k}\Omega$. This leads to a time constant of:

$$\tau_5 = 9.6 \times 10^{-9}\text{ s}$$

7.2.2 Emitter follower Buffer 1

DC Characterization

The emitter buffer was used between the first module and the gain cascode to keep the Req of the input pole of cascode low. This improves our input time constants with the high miller's capacitor.

The input resistance of this stage was expected to be 91 k Ω . It was kept high in order to keep the R_{eq} for the output pole of the previous cascode low. The collector current was expected to be 0.7mA and beta value was assumed to be 250 for this stage. With that the r_π and g_m calculated:

$$g_m = \frac{I_c}{V_t} = \frac{0.7 \text{ mA}}{26 \text{ mV}} = 0.0269 \text{ A/V}$$

$$r_\pi = \frac{\beta_0}{g_m} = \frac{250}{0.0269} = 9290 \text{ } \Omega$$

Since the R_{in} is expected to be 91k Ω and R_L is 22 k Ω , the biasing network chosen such that $R_{in} = R_{BIAS} \parallel [r_\pi + (1 + \beta_o)(R_E \parallel R_L)]$, while setting the V_E to 3.6V and V_B to 4.3V. The R_{BIAS} was calculated to around 100k Ω and with this the following equations were used to find the bias resistors.

$$\frac{9 - 4.26}{R_1} = \frac{4.26}{R_2} + 2.8 \times 10^{-6}$$
$$\left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = 100k\Omega$$

The R2 comes out to be approximately 200k Ω and R1 comes out to be approximately 200k Ω . Vb is 4.26V. With this bias resistors, the input resistance of the stage was calculated to be 88K Ω .

Midband Gain

With the selected component values, the midband gain of the stage was analyzed.

First, we calculate the equivalent resistances for the stage:

$$R_1 \parallel R_2 = 200 \text{ k}\Omega \parallel 200 \text{ k}\Omega = 100 \text{ k}\Omega$$

$$R_\pi = 9290 \text{ } \Omega$$

$$R_c \parallel R_L = 5.2 \text{ k}\Omega \parallel 22 \text{ k}\Omega = 4 \text{ k}\Omega$$

Next, we use overall voltage gain (A_{vo}) to calculate the gain:

$$A_v = \frac{(1 + \beta_o)(R_E \parallel R_L)}{r_\pi + (1 + \beta_o)(R_E \parallel R_L)}$$

$$A_v = \frac{(1 + 250)(5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega)}{9290\Omega + (1 + 250)(5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega)}$$

Finally, combining these gives the total gain: **-0.1dB**.

Frequency Response

With the selected component values, the pole frequencies of the gain stage was calculated. Since the poles from the beginning to input pole of the first cascodes are interactive, the time constants were calculated and the frequency poles will be calculated in the feedback calculation section.

$$\tau_1 = 3 \times 10^{-12} \times (1.3k \text{ k}\Omega \parallel 100 \text{ k}\Omega \parallel (9290 \text{ k}\Omega + (1 + 250) \times 4200\Omega)) = 3.85 \times 10^{-9}$$

$$\tau_2 = 36 \times 10^{-12} \times (9290 \text{ }\Omega \parallel \left[\frac{1.3 \text{ k}\Omega + 5.2k \text{ }\Omega \parallel 22k\Omega}{1 + (0.0269)(5.2k \text{ }\Omega \parallel 22k\Omega)} \right]) = 1.33 \times 10^{-9}$$

Overall, this gain stage achieves **-0.1dB** gain, and the bandwidth will be calculated in the later section.

7.2.3 Gain Cascode Stage 1

DC Characteristics

In order to use series/shunt feedback over the gain stages, a split emitter resistor was used by bypassing 1.3k Ω and leaving 100 Ω . This allows to increase the overall bandwidth of our gain stages without changing any of the DC biasing values. So, the DC values are same as the DC values of the other gain stage cascode. However, with 100 Ω the gain of the circuit decreases.

The input resistance (R_{in}) of this stage was set to be 7.5 k Ω because increasing the (R_{in}) will help us to have more gain. For this stage, the transistors operate with a transconductance g_m of 0.0346 A/V and a base-emitter resistance r_π of 8.6 k Ω . The collector current of the cascode was expected to 0.9mA.

$$g_m = \frac{I_c}{V_t} = \frac{0.9 \text{ mA}}{26 \text{ mV}} = 0.0346 \text{ A/V}$$

$$r_\pi = \frac{\beta_0}{g_m} = \frac{300}{0.0346} = 8600 \text{ } \Omega$$

Since R_{in} is expected to be ~ 22 k Ω , the biasing network must be chosen so that $r_\pi \parallel R_2 \parallel R_3 \approx 22$ k Ω . Additionally, R_2 and R_3 must be sized to bias the collector current (I_c) at approximately 1.26 V. The base current was calculated using the relationship between (I_c) and (I_b), which was 3 μ A.

Since the emitter resistor (R_e) is 1.4 k Ω in DC, it sets the emitter voltage (V_e) to 1.26 V. To forward-bias the NPN transistor, the base voltage (V_b) must be around 1.86 V to 1.96 V.

Using these parameters, the following equations were set up to solve for the exact values of R_1 , R_2 , and V_{b2} . The R_3 chosen to be 110 k Ω to ensure the bias network draws significantly more current than the calculated 3 μ A base current. This helps with overall DC power consumptions and staying under 12 mA budget. Also, the biasing resistor was picked to much larger than any AC resistors because we want DC biasing to be disappeared from the AC signal path of the amplifier.

$$\text{eq1} = \frac{1}{8666 + (301)(100)} + \frac{1}{110000} + \frac{1}{R_2} = \frac{1}{22000}$$

$$\text{eq2} = \frac{9 - V_{b2}}{R_1} = \frac{V_{b3} - 1.95}{110000} + 3 \times 10^{-6}$$

$$\text{eq3} = \frac{V_{b2} - 1.95}{110000} = \frac{1.95}{110000} + 3 \times 10^{-6}$$

The R_2 comes out to be approximately 110k Ω and R_1 comes out to be approximately 200k Ω . V_{b2} is 4.23V. With this bias resistors, the input resistance of the stage was calculated to be 22k Ω .

The selection of the collector resistor (R_c) was carefully, since it directly affects midband gain and high-frequency bandwidth. it was selected to be R_c of 3 k Ω . Because we have used emitter follower as buffer stage in order to fix the output pole. Since the input pole

of the emitter follower is large, small collector resistor was picked to keep the Req low such that output pole relatively large.

The parallel combination of these resistors ($R_c \parallel R_{load} \approx 2.9 \text{ k}\Omega$) sets the open-loop voltage gain. While increasing R_c would yield higher midband gain to help reach the 60 dB overall system goal, but it would also decrease output pole (f_3) to a lower frequency, failing our 7.2 MHz bandwidth requirement. Here is the DC operating point.

Midband Gain

With the selected component values, the midband gain of the stage was analyzed. First, we calculate the equivalent resistances for the stage:

$$\begin{aligned} R_3 \parallel R_2 &= 110 \text{ k}\Omega \parallel 110 \text{ k}\Omega = 55 \text{ k}\Omega \\ R_{in} &= 22 \text{ k}\Omega \\ R_c \parallel R_L &= 3 \text{ k}\Omega \parallel 88 \text{ k}\Omega = 2.9 \text{ k}\Omega \end{aligned}$$

Next, we break down the overall voltage gain (A_{vo}) into the individual stage gains (Common Emitter and Common Base). With R_E is added the gain equation changed:

$$A_v = - \left(\frac{R_{in}}{R_S + R_{in}} \right) \left[\frac{\beta_o(R_C \parallel R_L)}{r_{\pi 1} + (1 + \beta_o)R_E} \right]$$

Finally, plug in the values gives the total gain: **27 dB**.

$$\begin{aligned} A_v &= - \left(\frac{R_{in}}{R_S + R_{in}} \right) \left[\frac{\beta_{AC}(R_C \parallel R_6)}{r_{\pi} + (1 + \beta_{AC})R_4} \right] \\ A_v &= - \left(\frac{22 \text{ k}\Omega}{42 \text{ }\Omega + 22 \text{ k}\Omega} \right) \left[\frac{250(2.90 \text{ k}\Omega)}{33700 \text{ k}\Omega} \right] \\ G_v &\approx -21.47 \text{ V/V} \quad \boxed{26.6 \text{ dB}} \end{aligned}$$

$$20 \log_{10}(21.47) = 26.6 \text{ dB}$$

Frequency Response With the selected component values, the pole frequencies of the gain stage was calculated.

Since the poles from output poles of the first cascode to input pole of the second poles are interactive, the time constants were calculated and the frequency poles will be calculated in the feedback calculation section.

$$\tau_1 = 3.66 \times 10^{-12} \times (42 \Omega) = 1.5 \times 10^{-10}$$

$$\tau_2 = 13 \times 10^{-12} \times \left(8600 \Omega \parallel \left[\frac{40 \text{ k}\Omega + 100 \Omega}{1 + (0.0346)(100 \Omega)} \right] \right) = 1.27 \times 10^{-9}$$

$$\tau_3 = 56.5 \times 10^{-12} \times \left(\frac{1}{0.0346} \right) = 1.61 \times 10^{-9}$$

$$\tau_4 = 3 \times 10^{-12} \times (2.9 \text{ k}\Omega) = 8.7 \times 10^{-9}$$

Overall, this gain stage achieves **26.6dB** gain, and the bandwidth will be calculated in the later section.

7.2.4 Emitter follower Buffer 2

DC Characterization

Since connecting two cascode stages in cascade resulted in a poor output frequency pole, we introduced an emitter follower buffer in between to improve the frequency response. However, this causes the frequency poles to interact throughout the gain modules.

The input resistance of this stage was expected to be 88 k Ω . It was kept high in order to keep the R_{eq} for the output pole of the previous cascode low. The collector current was expected to be 0.7mA and beta value was assumed to be 250 for this stage. With that the r_π and g_m calculated:

$$g_m = \frac{I_c}{V_t} = \frac{0.7 \text{ mA}}{26 \text{ mV}} = 0.0269 \text{ A/V}$$

$$r_\pi = \frac{\beta_0}{g_m} = \frac{250}{0.0269} = 9290 \text{ } \Omega$$

Since the R_{in} is expected to be 88k Ω and R_L is 7.5 k Ω , the biasing network chosen such that $R_{in} = R_{BIAS} \parallel [r_\pi + (1 + \beta_o)(R_E \parallel R_L)]$, while setting the V_E to 3.6V and V_B to 4.3V. The R_{BIAS} was calculated to around 100k Ω and with this the following equations were used to find the bias resistors.

$$\frac{9 - 4.26}{R_1} = \frac{4.26}{R_2} + 2.8 \times 10^{-6}$$
$$\left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = 100k\Omega$$

The R_2 comes out to be approximately 200k Ω and R_1 comes out to be approximately 200k Ω . V_b is 4.26V. With this bias resistors, the input resistance of the stage was calculated to be 88K Ω .

Midband Gain

With the selected component values, the midband gain of the stage was analyzed. First, we calculate the equivalent resistances for the stage:

$$R_1 \parallel R_2 = 200 \text{ k}\Omega \parallel 200 \text{ k}\Omega = 100 \text{ k}\Omega$$

$$R_\pi = 9290 \text{ } \Omega$$

$$R_c \parallel R_L = 5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega = 3.07 \text{ k}\Omega$$

Next, we use overall voltage gain (A_{vo}) to calculate the gain:

$$A_v = \frac{(1 + \beta_o)(R_E \parallel R_L)}{r_\pi + (1 + \beta_o)(R_E \parallel R_L)}$$

$$A_v = \frac{(1 + 250)(5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega)}{9290\Omega + (1 + 250)(5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega)}$$

Finally, combining these gives the total gain: **-0.1dB**.

Frequency Response

With the selected component values, the pole frequencies of the gain stage was calculated. As it mentioned in the prior section, due to poles are interactive, the time constants were calculated and the frequency poles will be calculated in the feedback section.

$$\tau_1 = 3 \times 10^{-12} \times (3 \text{ k}\Omega \parallel 100 \text{ k}\Omega \parallel (9290 \text{ k}\Omega + (1 + 250) \times 3070\Omega)) = 8.97 \times 10^{-9}$$

$$\tau_2 = 36 \times 10^{-12} \times (9290 \text{ }\Omega \parallel \left[\frac{3 \text{ k}\Omega + 5.2\text{k } \Omega}{1 + (0.0269)(5.2\text{k } \Omega)} \right]) = 2.61 \times 10^{-9}$$

Overall, this gain stage achieves **-0.1dB** gain, and the bandwidth will be calculated in the later section.

7.2.5 Gain Cascode Stage 2

For this stage, same Cascode topology from previous stage. So, component values are identical except the R_{LOAD} and R_S , which are 7 k Ω and 37 Ω .

DC Characterization

The input resistance (R_{in}) of this stage was set to be 7.5 k Ω because increasing the (R_{in}) will help us to have more gain. For this stage, the transistors operate with a transconductance g_m of 0.0346 A/V and a base-emitter resistance r_π of 8.6 k Ω . The collector current of the cascode was expected to 0.9mA.

$$g_m = \frac{I_c}{V_t} = \frac{0.9 \text{ mA}}{26 \text{ mV}} = 0.0346 \text{ A/V}$$

$$r_\pi = \frac{\beta_0}{g_m} = \frac{300}{0.0346} = 8600 \text{ } \Omega$$

Since we are utilizing similar topology, same resistor values were chosen with similar intent as pervious stage. The chose resistor values are:

$$R_1 = 200000 \text{ } \Omega$$

$$R_2 = 110000 \text{ } \Omega$$

$$R_3 = 110000 \text{ } \Omega$$

$$R_c = 3000 \text{ } \Omega$$

$$R_e = 1400 \text{ } \Omega$$

Midband Gain

With the selected component values, the midband gain of the stage was calculated.

First, we calculate the equivalent resistances for the stage:

$$R_3 \parallel R_2 = 110 \text{ k}\Omega \parallel 110 \text{ k}\Omega = 55 \text{ k}\Omega$$

$$R_{in} = 7.5 \text{ } \Omega$$

$$R_c \parallel R_L = 3 \text{ k}\Omega \parallel 7 \text{ k}\Omega = 2.1 \text{ k}\Omega$$

Next, we break down the overall voltage gain (A_{vo}) into the individual stage gains (Common Emitter and Common Base):

$$A_{vo} = \frac{v_o}{v_2} \cdot \frac{v_2}{v_1} \cdot \frac{v_1}{v_s}$$

$$\frac{v_1}{v_s} = \frac{R_{in}}{R_s + R_{in}}$$

$$\frac{v_2}{v_1} \Rightarrow \text{CE amp} : -g_{m1}R_x \approx \frac{-g_{m1}}{g_{m2}} \approx -1$$

$$\frac{v_o}{v_2} \Rightarrow \text{CB amp} : \frac{R_c \parallel R_L}{R_s + 1/g_m} \approx g_m(R_c \parallel R_L)$$

Finally, combining these gives the total gain: **37.18dB**.

$$\begin{aligned}
 A_{vo} &= \left(\frac{R_{in}}{R_s + R_{in}} \right) \cdot \left[-g_m(R_c \parallel R_L) \right] \\
 &= \left(\frac{7500}{36 + 7500} \right) \times (-0.0346) \times 2.1 \text{ k}\Omega \\
 &= -72.31 \text{ V/V} \quad \text{or} \quad \boxed{37.18 \text{ dB}}
 \end{aligned}$$

$$20 \log_{10}(72.31) = 37.18 \text{ dB}$$

Frequency Response

With the selected component values, the pole frequencies of the gain stage was calculated. Since the every frequency poles of in circuit are interactive, only time constants were calculated and they will be used in the later section to find the bandwidth.

$$\tau_1 = 46 \times 10^{-12} \times (36\Omega \parallel 55 \text{ k}\Omega \parallel 8600 \text{ k}\Omega) = 1.65 \times 10^{-9}$$

$$\tau_2 = 3 \times 10^{-12} \times \left(\frac{1}{0.0346} \right) = 8.67 \times 10^{-11}$$

$$\tau_3 = 3 \times 10^{-12} \times (2.1 \text{ k}\Omega) = 6.3 \times 10^{-9}$$

Overall, this gain stage achieves **37.18 dB** gain, and the bandwidth will be calculated in the later section.

7.2.6 Darlington Output Stage

DC Characteristics

Though research of typical darlington configurations we decided to implement an RE bias resistor to stabilize our first values r_π to reduce the effective dominance of variations from β . This required us to bias the first emitter follower at 1 mA, and subsequently the second emitter follower at approximately 0.35mA. This happened to be the ideal spot through repeated hand calculations for a very stable Rout value and acceptable gain loss. We used the relationships defined in the table below to derive our resistor values and biases.

R_{out} Characteristics The priority of our output stage was providing a low output impedance ($75 \Omega \pm 20\%$). Our latest design accomplishes this goal in an impressively precise manner. The R_{out} equation for an emitter follower is defined as:

$$R_{out} = R_E \parallel \frac{r_\pi + R_S \parallel R_{BIAS}}{(1 + \beta_o)}$$

In our design we calculated our first emitter follower to have an Rout of about 46Ω and our second the true output to have an value of approximately 76Ω . This value depending on beta variations ranges from 71-79, staying well within our specification target.

The worst-case total system current/power draw of the system is 60.3 mW.

Midband Gain The midband gain of an emitter follower is defined by:

$$A_v = \frac{(1 + \beta_o)(R_E \parallel R_L)}{r_\pi + (1 + \beta_o)(R_E \parallel R_L)}$$

In our design this stage features essentially two emitter followers stacked so we utilized the equation above for each and multiplied them together at our nominal operating point yields $A_v = 0.49$, which corresponds to a gain of **-6.2 dB**.

Every pole in this amplifier is interactive so only the time constants will be calculated in this section. The input pole combines the output capacitance from the preceding cascode with C_μ from this transistor, yielding $C_{eq} = 5$ pF and a resistance of $2.1\text{k}\Omega$. This produces a time constant of:

$$\tau_1 = 1.05 \times 10^{-8} \text{ s}$$

The second pole is determined by $C_{\pi 1} = 0.6$ pF facing a resistance of approximately 54Ω , giving:

$$\tau_2 = 3.24 \times 10^{-11} \text{ s}$$

The next pole is $c_{\mu 2}$ which has an equivalent resistance of 49Ω . This produces:

$$\tau_3 = 4.9 \times 10^{-11} \text{ s}$$

Lastly we have $c_{\pi 2}$ which possess an equivalent resistance of 11Ω yielding a time constant of

$$\tau_4 = 6.6 \times 10^{-12} \text{ s}$$

Combining both time constants, the resulting pole frequency is:

$$f = \frac{1}{2\pi(\tau_1 + \tau_2)} = \frac{1}{2\pi(4.095 \times 10^{-9} + 1.926 \times 10^{-8})} = 7.53 \text{ MHz}$$

This means the emitter follower output stage is within specifications.

7.2.7 Feedback calculation for gain stage

While our design for our gain stage was solid in terms of frequency response, our gain was very susceptible to β_o variations and also out of specification. We decided to approach feedback to mitigate these issues with a series/shunt feedback topology extending over the entire gain stage. First we must address the calculations for our open loop frequency response:

As it was calculated in the input section, the first pole frequency was calculated to be 9.98MHz and the last pole frequency was calculated to be 15.3MHz. The rest is calculated here.

$$\begin{aligned} f_2 &= \frac{1}{2\pi(8.32 \times 10^{-9} + 6 \times 10^{-10} + 3.78 \times 10^{-12} + 4.13 \times 10^{-10})} \\ &= 17\text{MHz} \\ f_3 &= \frac{1}{2\pi(1.45 \times 10^{-8} + 7.56 \times 10^{-10} + 7.92 \times 10^{-10})} \\ &= 9.92\text{MHz} \end{aligned}$$

The f_2 and f_3 are the poles in the open loop and since neither of the poles are dominating, we used the following formula to find the bandwidth of the loop.

$$f_H = 1 / \sqrt{\frac{1}{f_{P1}^2} + \frac{1}{f_{P2}^2}} = 8.5\text{MHz}$$

We started our topology design with the series resistor value and utilized that to find our shunt resistor value. As a practical starting point, we chose our R_E value as 100 Ω in a splitter resistor format to maintain our biasing. Next we began our open loop gain calculations including the loading effect from the feedback resistors. This manifests as R_E in parallel with R_F at the emitter of the transistor in the gain stage and $R_F + R_E$ in parallel with the load at the output of gain stage.

Open Loop Gain

Open loop gain Calculations for the gain stage seen [here](#):

$$\frac{v_i}{v_s} = \frac{22 \text{ k}\Omega}{42 \text{ }\Omega + 22 \text{ k}\Omega} = 0.9999$$

$$\frac{v_1}{v_i} = \frac{-\beta_o(R_C \parallel R_L)}{r_\pi + (1 + \beta_o)R_E} = \frac{-300 \left(\frac{36 \text{ k}\Omega}{301}\right) 100 \text{ }\Omega}{8.6 \text{ k}\Omega + 301(100 \text{ }\Omega)} = -0.221$$

$$\frac{v_2}{v_1} = g_m(R_C \parallel R_L) = 34.6 \text{ mA/V} \left(3 \text{ k}\Omega \parallel (9.3 \text{ k}\Omega + 301(55 \text{ k}\Omega \parallel 8.6 \text{ k}\Omega)) \right) = 103.7$$

$$\frac{v_3}{v_2} = \frac{(1 + \beta_o)(R_E \parallel R_L)}{r_\pi + (1 + \beta_o)(R_E \parallel R_L)} = \frac{301(5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega)}{9.3 \text{ k}\Omega + 301(5.2 \text{ k}\Omega \parallel 7.5 \text{ k}\Omega)} = 0.996$$

$$\frac{v_4}{v_3} = -g_m(R_C \parallel R_L) = -34.6 \text{ mA/V} \left(\frac{8.6 \text{ k}\Omega}{301} \right) = -0.998$$

$$\frac{v_o}{v_4} = g_m(R_C \parallel R_L) = 34.6 \text{ mA/V} (3 \text{ k}\Omega \parallel 7 \text{ k}\Omega) = 72.24$$

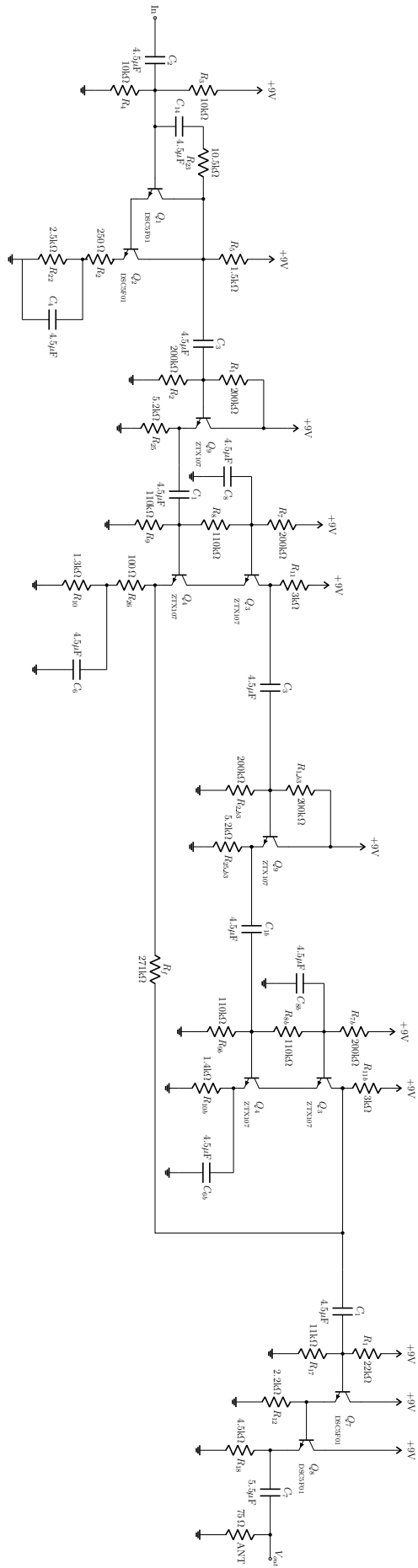
$$A_v = \left(\frac{v_i}{v_s} \right) \left(\frac{v_1}{v_i} \right) \left(\frac{v_2}{v_1} \right) \left(\frac{v_3}{v_2} \right) \left(\frac{v_4}{v_3} \right) \left(\frac{v_o}{v_4} \right) = 1612.8 = 64.2 \text{ dB}$$

After determining our gain was roughly 64dB we selected our R_F value on the basis of a relationship where:

$$1000 = \frac{A}{1 + A\beta} = \frac{64 \text{ dB}}{1 + 64 \text{ dB} \left(\frac{100}{100 + R_F} \right)}$$

This relationship relates our open loop gain to our desired gain of 60dB or 1000 V/V. Using this relationship we determined our β factor to be $\frac{1}{2701}$ and the value of R_F to be 271 k Ω . This also extends our 3dB point further by the feedback factor which is $(1 + A\beta) = 1.586$. Our new final 3dB point is at 13.481 MHz

7.3 Appendix C / Full Feedback Schematic



7.4 Appendix D / Cost Analysis

Here is the table that shows the cost for each available transistors.

Device	Use	Cost
Q2N2222	Gen Purpose	\$0.50
TIS98	Low Current	\$0.75
ZTX107	Low Current	\$1.00
Q2N2222-P	Gen Purpose	\$0.50
TIS98-P	Low Current	\$0.75
ZTX107-P	Low Current	\$1.00
DSC7501	Low VCE	\$1.25

Here is the cost table for other components in the circuit.

Components	Cost
Resistor	\$0.10
Capacitor	\$0.20

For the power cost, there is a power profit penalty (cost) of \$0.15 per mW over 60mW.

Component	Count	Cost Per Components (\$)	Total Cost (\$)
ZTX107	6	1.00	6.00
DSC5F01	4	1.25	5.00
Resistor	29	0.10	2.90
Capacitor	14	0.20	2.80
Total			16.70

The total cost of the design calculated to be **\$16.70**, and there was no power profit penalty.

7.5 Appendix E / Operating points(Simulated)

7.5.1 Input Stage transistors

The input stage has two DSC5F01 transistors and their operating points are shown at -LO,-NOM, and -HI simulations.

--- Bipolar Transistors ---		
Name:	Q1	Q2
Model:	dsc5f01-lo	dsc5f01-lo
Ib:	6.84e-07	2.33e-05
Ic:	2.26e-05	1.13e-03
Vbe:	6.16e-01	7.17e-01
Vbc:	-2.78e+00	-3.40e+00
Vce:	3.40e+00	4.11e+00
BetaDC:	3.30e+01	4.85e+01
Gm:	8.71e-04	4.32e-02
Rpi:	4.57e+04	1.16e+03
Rx:	5.20e-01	5.20e-01
Ro:	9.98e+05	2.05e+04
Cbe:	1.90e-12	5.57e-12
Cbc:	1.26e-12	1.23e-12
Cjs:	0.00e+00	0.00e+00
BetaAC:	3.98e+01	5.01e+01
Cbx:	0.00e+00	0.00e+00
Ft:	4.38e+07	1.01e+09

Figure 14: -LO Operating point for DSC5F01s.

--- Bipolar Transistors ---		
Name:	Q1	Q2
Model:	dsc5f01-nom	dsc5f01-nom
Ib:	2.90e-07	1.23e-05
Ic:	1.20e-05	1.14e-03
Vbe:	6.00e-01	7.18e-01
Vbc:	-2.77e+00	-3.37e+00
Vce:	3.37e+00	4.08e+00
BetaDC:	4.15e+01	9.28e+01
Gm:	4.65e-04	4.39e-02
Rpi:	1.26e+05	2.27e+03
Rx:	5.20e-01	5.20e-01
Ro:	1.87e+06	2.02e+04
Cbe:	1.80e-12	5.62e-12
Cbc:	1.26e-12	1.23e-12
Cjs:	0.00e+00	0.00e+00
BetaAC:	5.86e+01	9.96e+01
Cbx:	0.00e+00	0.00e+00
Ft:	2.42e+07	1.02e+09

Figure 15: -NOM Operating point for DSC5F01s.

```

          --- Bipolar Transistors ---
Name:      Q1      Q2
Model:     dsc5f01-hi dsc5f01-hi
Ib:        2.03e-07  8.85e-06
Ic:        8.65e-06  1.15e-03
Vbe:       5.91e-01  7.18e-01
Vbc:       -2.76e+00 -3.35e+00
Vce:       3.35e+00  4.07e+00
BetaDC:    4.26e+01  1.30e+02
Gm:        3.34e-04  4.42e-02
Rpi:       1.96e+05  3.26e+03
Rx:        5.20e-01  5.20e-01
Ro:        2.60e+06  2.00e+04
Cbe:       1.76e-12  5.64e-12
Cbc:       1.26e-12  1.23e-12
Cjs:       0.00e+00  0.00e+00
BetaAC:    6.55e+01  1.44e+02
Cbx:       0.00e+00  0.00e+00
Ft:        1.76e+07  1.02e+09

```

Figure 16: -HI Operating point for DSC5F01s.

7.5.2 Emitter Follower Buffer 1 transistor

The emitter follower buffer has one ZTX107 transistor and its operating points are shown at -LO,-NOM, and -HI simulations.

```

- --- Bipolar Transistors ---
Name:      Q1
Model:     ztx107-lo
Ib:        3.49e-06
Ic:        6.74e-04
Vbe:       6.27e-01
Vbc:       -4.85e+00
Vce:       5.48e+00
BetaDC:    1.93e+02
Gm:        2.60e-02
Rpi:       7.97e+03
Rx:        5.60e-01
Ro:        1.20e+05
Cbe:       3.66e-11
Cbc:       1.87e-12
Cjs:       0.00e+00
BetaAC:    2.07e+02
Cbx:       0.00e+00
Ft:        1.08e+08
```

Figure 17: -LO Operating point for ZTX107.

```

- --- Bipolar Transistors ---
Name:      Q1
Model:     ztx107-nom
Ib:        2.55e-06
Ic:        6.93e-04
Vbe:       6.28e-01
Vbc:       -4.75e+00
Vce:       5.38e+00
BetaDC:    2.72e+02
Gm:        2.67e-02
Rpi:       1.13e+04
Rx:        5.60e-01
Ro:        1.16e+05
Cbe:       3.71e-11
Cbc:       1.88e-12
Cjs:       0.00e+00
BetaAC:    3.01e+02
Cbx:       0.00e+00
Ft:        1.09e+08
```

Figure 18: -NOM Operating point for ZTX107.

```

--- Bipolar Transistors ---
Name:      Q1
Model:     ztx107-hi
Ib:        1.84e-06
Ic:        7.07e-04
Vbe:       6.28e-01
Vbc:       -4.68e+00
Vce:       5.31e+00
BetaDC:    3.84e+02
Gm:        2.73e-02
Rpi:       1.63e+04
Rx:        5.60e-01
Ro:        1.14e+05
Cbe:       3.74e-11
Cbc:       1.89e-12
Cjs:       0.00e+00
BetaAC:    4.45e+02
Cbx:       0.00e+00
Ft:        1.10e+08

```

Figure 19: -HI Operating point for ZTX107.

7.5.3 Gain stage transistors

The gain stage consists of two cascodes, emitter follower buffer, and feedback. The schematic without the feedback can be seen here. In total, it has 5 transistors and here is the operating points at different simulations. All of them are ZTX107.

--- Bipolar Transistors ---					
Name:	Q1	Q2	Q3	Q4	Q5
Model:	ztx107-lo	ztx107-lo	ztx107-lo	ztx107-lo	ztx107-lo
Ib:	4.27e-06	4.33e-06	4.27e-06	4.33e-06	3.49e-06
Ic:	8.12e-04	8.16e-04	8.12e-04	8.16e-04	6.74e-04
Vbe:	6.33e-01	6.33e-01	6.33e-01	6.33e-01	6.27e-01
Vbc:	-2.52e+00	-1.63e+00	-2.52e+00	-1.63e+00	-4.85e+00
Vce:	3.16e+00	2.26e+00	3.16e+00	2.26e+00	5.48e+00
BetaDC:	1.90e+02	1.88e+02	1.90e+02	1.88e+02	1.93e+02
Gm:	3.13e-02	3.14e-02	3.13e-02	3.14e-02	2.60e-02
Rpi:	6.49e+03	6.39e+03	6.49e+03	6.39e+03	7.97e+03
Rx:	5.60e-01	5.60e-01	5.60e-01	5.60e-01	5.60e-01
Ro:	9.67e+04	9.51e+04	9.67e+04	9.51e+04	1.20e+05
Cbe:	4.01e-11	4.02e-11	4.01e-11	4.02e-11	3.66e-11
Cbc:	2.26e-12	2.53e-12	2.26e-12	2.53e-12	1.87e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	2.03e+02	2.01e+02	2.03e+02	2.01e+02	2.07e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.18e+08	1.17e+08	1.18e+08	1.17e+08	1.08e+08

Figure 20: -LO Operating point for ZTX107s. Q1 and Q2 are cascode 1 transistors. Q5 is emitter follower. Q3 and Q4 are cascode 2 transistors.

--- Bipolar Transistors ---					
Name:	Q1	Q2	Q3	Q4	Q5
Model:	ztx107-nom	ztx107-nom	ztx107-nom	ztx107-nom	ztx107-nom
Ib:	3.34e-06	3.37e-06	3.34e-06	3.37e-06	2.55e-06
Ic:	9.03e-04	9.06e-04	9.03e-04	9.06e-04	6.93e-04
Vbe:	6.36e-01	6.36e-01	6.36e-01	6.36e-01	6.28e-01
Vbc:	-2.10e+00	-1.64e+00	-2.10e+00	-1.64e+00	-4.75e+00
Vce:	2.74e+00	2.28e+00	2.74e+00	2.28e+00	5.38e+00
BetaDC:	2.70e+02	2.69e+02	2.70e+02	2.69e+02	2.72e+02
Gm:	3.47e-02	3.49e-02	3.47e-02	3.49e-02	2.67e-02
Rpi:	8.54e+03	8.46e+03	8.54e+03	8.46e+03	1.13e+04
Rx:	5.60e-01	5.60e-01	5.60e-01	5.60e-01	5.60e-01
Ro:	8.65e+04	8.56e+04	8.65e+04	8.56e+04	1.16e+05
Cbe:	4.23e-11	4.24e-11	4.23e-11	4.24e-11	3.71e-11
Cbc:	2.37e-12	2.52e-12	2.37e-12	2.52e-12	1.88e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	2.96e+02	2.95e+02	2.96e+02	2.95e+02	3.01e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.24e+08	1.23e+08	1.24e+08	1.23e+08	1.09e+08

Figure 21: -NOM Operating point for ZTX107s. Q1 and Q2 are cascode 1 transistors. Q5 is emitter follower. Q3 and Q4 are cascode 2 transistors.

--- Bipolar Transistors ---					
Name:	Q1	Q2	Q3	Q4	Q5
Model:	ztx107-hi	ztx107-hi	ztx107-hi	ztx107-hi	ztx107-hi
Ib:	2.53e-06	2.54e-06	2.53e-06	2.54e-06	1.84e-06
Ic:	9.81e-04	9.84e-04	9.81e-04	9.84e-04	7.07e-04
Vbe:	6.38e-01	6.38e-01	6.38e-01	6.38e-01	6.28e-01
Vbc:	-1.74e+00	-1.66e+00	-1.74e+00	-1.66e+00	-4.68e+00
Vce:	2.38e+00	2.30e+00	2.38e+00	2.30e+00	5.31e+00
BetaDC:	3.88e+02	3.88e+02	3.88e+02	3.88e+02	3.84e+02
Gm:	3.77e-02	3.78e-02	3.77e-02	3.78e-02	2.73e-02
Rpi:	1.18e+04	1.17e+04	1.18e+04	1.17e+04	1.63e+04
Rx:	5.60e-01	5.60e-01	5.60e-01	5.60e-01	5.60e-01
Ro:	7.91e+04	7.89e+04	7.91e+04	7.89e+04	1.14e+05
Cbe:	4.43e-11	4.43e-11	4.43e-11	4.43e-11	3.74e-11
Cbc:	2.49e-12	2.52e-12	2.49e-12	2.52e-12	1.89e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	4.45e+02	4.44e+02	4.45e+02	4.44e+02	4.45e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.28e+08	1.29e+08	1.28e+08	1.29e+08	1.10e+08

Figure 22: -HI Operating point for ZTX107s. Q1 and Q2 are cascode 1 transistors. Q5 is emitter follower. Q3 and Q4 are cascode 2 transistors.

7.5.4 Output Stage transistors

The output stage has two DSC5F01 transistors and their operating points are shown at -LO,-NOM, and -HI simulations. Ignore Q10.

Name:	Q10	Q1	Q2
Model:	ztx107-lo	dsc5f01-lo	dsc5f01-lo
Ib:	3.49e-06	1.80e-05	6.20e-06
Ic:	6.74e-04	9.69e-04	3.22e-04
Vbe:	6.27e-01	7.11e-01	6.81e-01
Vbc:	-4.85e+00	-6.13e+00	-6.84e+00
Vce:	5.48e+00	6.84e+00	7.52e+00
BetaDC:	1.93e+02	5.37e+01	5.19e+01
Gm:	2.60e-02	3.72e-02	1.24e-02
Rpi:	7.97e+03	1.50e+03	4.49e+03
Rx:	5.60e-01	5.20e-01	5.20e-01
Ro:	1.20e+05	2.67e+04	8.26e+04
Cbe:	3.66e-11	5.09e-12	3.12e-12
Cbc:	1.87e-12	1.13e-12	1.11e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00
BetaAC:	2.07e+02	5.58e+01	5.57e+01
Cbx:	0.00e+00	0.00e+00	0.00e+00
Ft:	1.08e+08	9.52e+08	4.67e+08

Figure 23: -LO Operating point for DSC5F01s.

Name:	Q10	Q1	Q2
Model:	ztx107-nom	dsc5f01-nom	dsc5f01-nom
Ib:	2.55e-06	9.83e-06	3.60e-06
Ic:	6.93e-04	1.00e-03	3.37e-04
Vbe:	6.28e-01	7.11e-01	6.82e-01
Vbc:	-4.75e+00	-6.07e+00	-6.78e+00
Vce:	5.38e+00	6.78e+00	7.47e+00
BetaDC:	2.72e+02	1.02e+02	9.37e+01
Gm:	2.67e-02	3.85e-02	1.30e-02
Rpi:	1.13e+04	2.87e+03	8.21e+03
Rx:	5.60e-01	5.20e-01	5.20e-01
Ro:	1.16e+05	2.58e+04	7.86e+04
Cbe:	3.71e-11	5.19e-12	3.17e-12
Cbc:	1.88e-12	1.13e-12	1.11e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00
BetaAC:	3.01e+02	1.10e+02	1.07e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00
Ft:	1.09e+08	9.69e+08	4.83e+08

Figure 24: -NOM Operating point for DSC5F01s.

Name:	Q10	Q1	Q2
Model:	ztx107-hi	dsc5f01-hi	dsc5f01-hi
Ib:	1.84e-06	7.14e-06	2.74e-06
Ic:	7.07e-04	1.01e-03	3.42e-04
Vbe:	6.28e-01	7.12e-01	6.83e-01
Vbc:	-4.68e+00	-6.05e+00	-6.76e+00
Vce:	5.31e+00	6.76e+00	7.45e+00
BetaDC:	3.84e+02	1.42e+02	1.25e+02
Gm:	2.73e-02	3.89e-02	1.32e-02
Rpi:	1.63e+04	4.09e+03	1.13e+04
Rx:	5.60e-01	5.20e-01	5.20e-01
Ro:	1.14e+05	2.55e+04	7.74e+04
Cbe:	3.74e-11	5.22e-12	3.19e-12
Cbc:	1.89e-12	1.13e-12	1.12e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00
BetaAC:	4.45e+02	1.59e+02	1.50e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00
Ft:	1.10e+08	9.74e+08	4.89e+08

Figure 25: -HI Operating point for DSC5F01s.

7.5.5 Full Circuit Operating Points

In total there are 10 transistors. Their operating points were simulated at -LO,-NOM, and -HI.

--- Bipolar Transistors ---					
Name:	Q3	Q4	Q5	Q6	Q9
Model:	ztx107-lo	ztx107-lo	ztx107-lo	ztx107-lo	ztx107-lo
Ib:	4.27e-06	4.33e-06	4.27e-06	4.33e-06	3.49e-06
Ic:	8.12e-04	8.16e-04	8.12e-04	8.16e-04	6.74e-04
Vbe:	6.33e-01	6.33e-01	6.33e-01	6.33e-01	6.27e-01
Vbc:	-2.52e+00	-1.63e+00	-2.52e+00	-1.63e+00	-4.85e+00
Vce:	3.16e+00	2.26e+00	3.16e+00	2.26e+00	5.48e+00
BetaDC:	1.90e+02	1.88e+02	1.90e+02	1.88e+02	1.93e+02
Gm:	3.13e-02	3.14e-02	3.13e-02	3.14e-02	2.60e-02
Rpi:	6.49e+03	6.39e+03	6.49e+03	6.39e+03	7.97e+03
Rx:	5.60e-01	5.60e-01	5.60e-01	5.60e-01	5.60e-01
Ro:	9.67e+04	9.51e+04	9.67e+04	9.51e+04	1.20e+05
Cbe:	4.01e-11	4.02e-11	4.01e-11	4.02e-11	3.66e-11
Cbc:	2.26e-12	2.53e-12	2.26e-12	2.53e-12	1.87e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	2.03e+02	2.01e+02	2.03e+02	2.01e+02	2.07e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.18e+08	1.17e+08	1.18e+08	1.17e+08	1.08e+08

Name:	Q10	Q1	Q2	Q7	Q8
Model:	ztx107-lo	dsc5f01-lo	dsc5f01-lo	dsc5f01-lo	dsc5f01-lo
Ib:	3.49e-06	1.80e-05	6.20e-06	6.84e-07	2.33e-05
Ic:	6.74e-04	9.69e-04	3.22e-04	2.26e-05	1.13e-03
Vbe:	6.27e-01	7.11e-01	6.81e-01	6.16e-01	7.17e-01
Vbc:	-4.85e+00	-6.13e+00	-6.84e+00	-2.78e+00	-3.40e+00
Vce:	5.48e+00	6.84e+00	7.52e+00	3.40e+00	4.11e+00
BetaDC:	1.93e+02	5.37e+01	5.19e+01	3.30e+01	4.85e+01
Gm:	2.60e-02	3.72e-02	1.24e-02	8.71e-04	4.32e-02
Rpi:	7.97e+03	1.50e+03	4.49e+03	4.57e+04	1.16e+03
Rx:	5.60e-01	5.20e-01	5.20e-01	5.20e-01	5.20e-01
Ro:	1.20e+05	2.67e+04	8.26e+04	9.98e+05	2.05e+04
Cbe:	3.66e-11	5.09e-12	3.12e-12	1.90e-12	5.57e-12
Cbc:	1.87e-12	1.13e-12	1.11e-12	1.26e-12	1.23e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	2.07e+02	5.58e+01	5.57e+01	3.98e+01	5.01e+01
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.08e+08	9.52e+08	4.67e+08	4.38e+07	1.01e+09

Figure 26: -LO Operating point for DSC5F01s.

--- Bipolar Transistors ---					
Name:	Q3	Q4	Q5	Q6	Q9
Model:	ztx107-nom	ztx107-nom	ztx107-nom	ztx107-nom	ztx107-nom
Ib:	3.34e-06	3.37e-06	3.34e-06	3.37e-06	2.55e-06
Ic:	9.03e-04	9.06e-04	9.03e-04	9.06e-04	6.93e-04
Vbe:	6.36e-01	6.36e-01	6.36e-01	6.36e-01	6.28e-01
Vbc:	-2.10e+00	-1.64e+00	-2.10e+00	-1.64e+00	-4.75e+00
Vce:	2.74e+00	2.28e+00	2.74e+00	2.28e+00	5.38e+00
BetaDC:	2.70e+02	2.69e+02	2.70e+02	2.69e+02	2.72e+02
Gm:	3.47e-02	3.49e-02	3.47e-02	3.49e-02	2.67e-02
Rpi:	8.54e+03	8.46e+03	8.54e+03	8.46e+03	1.13e+04
Rx:	5.60e-01	5.60e-01	5.60e-01	5.60e-01	5.60e-01
Ro:	8.65e+04	8.56e+04	8.65e+04	8.56e+04	1.16e+05
Cbe:	4.23e-11	4.24e-11	4.23e-11	4.24e-11	3.71e-11
Cbc:	2.37e-12	2.52e-12	2.37e-12	2.52e-12	1.88e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	2.96e+02	2.95e+02	2.96e+02	2.95e+02	3.01e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.24e+08	1.23e+08	1.24e+08	1.23e+08	1.09e+08

Name:	Q10	Q1	Q2	Q7	Q8
Model:	ztx107-nom	dsc5f01-nom	dsc5f01-nom	dsc5f01-nom	dsc5f01-nom
Ib:	2.55e-06	9.83e-06	3.60e-06	2.90e-07	1.23e-05
Ic:	6.93e-04	1.00e-03	3.37e-04	1.20e-05	1.14e-03
Vbe:	6.28e-01	7.11e-01	6.82e-01	6.00e-01	7.18e-01
Vbc:	-4.75e+00	-6.07e+00	-6.78e+00	-2.77e+00	-3.37e+00
Vce:	5.38e+00	6.78e+00	7.47e+00	3.37e+00	4.08e+00
BetaDC:	2.72e+02	1.02e+02	9.37e+01	4.15e+01	9.28e+01
Gm:	2.67e-02	3.85e-02	1.30e-02	4.65e-04	4.39e-02
Rpi:	1.13e+04	2.87e+03	8.21e+03	1.26e+05	2.27e+03
Rx:	5.60e-01	5.20e-01	5.20e-01	5.20e-01	5.20e-01
Ro:	1.16e+05	2.58e+04	7.86e+04	1.87e+06	2.02e+04
Cbe:	3.71e-11	5.19e-12	3.17e-12	1.80e-12	5.62e-12
Cbc:	1.88e-12	1.13e-12	1.11e-12	1.26e-12	1.23e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	3.01e+02	1.10e+02	1.07e+02	5.86e+01	9.96e+01
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.09e+08	9.69e+08	4.83e+08	2.42e+07	1.02e+09

Figure 27: -NOM Operating point for DSC5F01s.

--- Bipolar Transistors ---					
Name:	Q3	Q4	Q5	Q6	Q9
Model:	ztx107-hi	ztx107-hi	ztx107-hi	ztx107-hi	ztx107-hi
Ib:	2.53e-06	2.54e-06	2.53e-06	2.54e-06	1.84e-06
Ic:	9.81e-04	9.84e-04	9.81e-04	9.84e-04	7.07e-04
Vbe:	6.38e-01	6.38e-01	6.38e-01	6.38e-01	6.28e-01
Vbc:	-1.74e+00	-1.66e+00	-1.74e+00	-1.66e+00	-4.68e+00
Vce:	2.38e+00	2.30e+00	2.38e+00	2.30e+00	5.31e+00
BetaDC:	3.88e+02	3.88e+02	3.88e+02	3.88e+02	3.84e+02
Gm:	3.77e-02	3.78e-02	3.77e-02	3.78e-02	2.73e-02
Rpi:	1.18e+04	1.17e+04	1.18e+04	1.17e+04	1.63e+04
Rx:	5.60e-01	5.60e-01	5.60e-01	5.60e-01	5.60e-01
Ro:	7.91e+04	7.89e+04	7.91e+04	7.89e+04	1.14e+05
Cbe:	4.43e-11	4.43e-11	4.43e-11	4.43e-11	3.74e-11
Cbc:	2.49e-12	2.52e-12	2.49e-12	2.52e-12	1.89e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	4.45e+02	4.44e+02	4.45e+02	4.44e+02	4.45e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.28e+08	1.29e+08	1.28e+08	1.29e+08	1.10e+08

Name:	Q10	Q1	Q2	Q7	Q8
Model:	ztx107-hi	dsc5f01-hi	dsc5f01-hi	dsc5f01-hi	dsc5f01-hi
Ib:	1.84e-06	7.14e-06	2.74e-06	2.03e-07	8.85e-06
Ic:	7.07e-04	1.01e-03	3.42e-04	8.65e-06	1.15e-03
Vbe:	6.28e-01	7.12e-01	6.83e-01	5.91e-01	7.18e-01
Vbc:	-4.68e+00	-6.05e+00	-6.76e+00	-2.76e+00	-3.35e+00
Vce:	5.31e+00	6.76e+00	7.45e+00	3.35e+00	4.07e+00
BetaDC:	3.84e+02	1.42e+02	1.25e+02	4.26e+01	1.30e+02
Gm:	2.73e-02	3.89e-02	1.32e-02	3.34e-04	4.42e-02
Rpi:	1.63e+04	4.09e+03	1.13e+04	1.96e+05	3.26e+03
Rx:	5.60e-01	5.20e-01	5.20e-01	5.20e-01	5.20e-01
Ro:	1.14e+05	2.55e+04	7.74e+04	2.60e+06	2.00e+04
Cbe:	3.74e-11	5.22e-12	3.19e-12	1.76e-12	5.64e-12
Cbc:	1.89e-12	1.13e-12	1.12e-12	1.26e-12	1.23e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	4.45e+02	1.59e+02	1.50e+02	6.55e+01	1.44e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	1.10e+08	9.74e+08	4.89e+08	1.76e+07	1.02e+09

Figure 28: -HI Operating point for DSC5F01s.